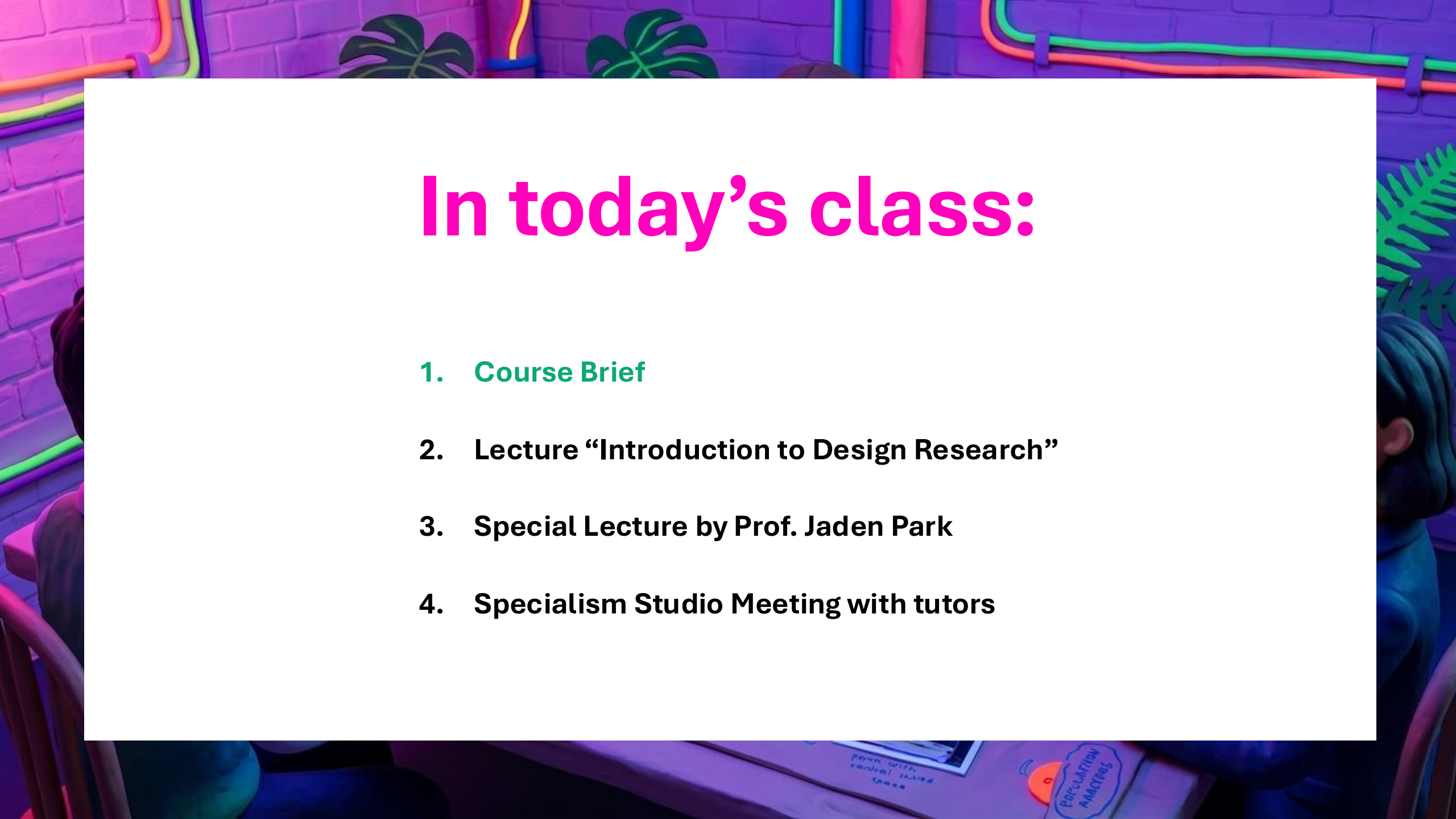


SD5008
Research and Analysis for Design

WEEK 01

Jen Yoohyun LEE / Henry Rui HE / Mohana DAS / Bharati DAS



In today’s class:

1. Course Brief
2. Lecture “Introduction to Design Research”
3. Special Lecture by Prof. Jaden Park
4. Specialism Studio Meeting with tutors

Synopsis

Design fundamentally aims to improve human experience, yet uncritical interventions risk causing unintended ecological, social, or systemic harm.

To ensure design is contextually responsive and deeply critical of its impacts, this course positions research as the foundational premise for a rigorous, professional practice.

Unlike scientific or humanities research — which focus on observing reality or testing hypotheses — design research is an applied, process-oriented endeavor that actively shapes positive change through human-centered action and material intervention.

By learning to observe, listen, measure, explore, and collaborate with stakeholders, you will gain the essential skills needed to navigate complex contexts, uncover deep insights, and identify opportunities for meaningful design interventions.

Course Components

The course includes three components:

- **Structured content lectures** in which the tutors will discuss topics related to design research and analysis for students' information and guidance in developing research.

- **Research study projects** in which groups of students will design and conduct a short study that includes a literature review, data collection, analysis, and communication. These projects must include a component of first-hand data gathering and/or generation, and the progress is to be demonstrated and communicated through presentations and the final report.

- **In-class individual written assessment** in which students will individually reflect on and articulate their in-depth understanding of research design at the end of the semester.

Presentations

- A total of 5 iteratively developing presentations throughout the semester
- Student groups present their progress on their research study project.
- 5 ~ 15 minutes per group
- Visual, in-person presentation (.pdf or .ppt) plus verbal presentation.
- Must include written, visual, and verbal content developed to a professional standard.

Research Study Report [Due 6 December]

- A written document of approximately 5,000-8,000 words, in English
- Submitted to plagiarism checking with report appended
- Must include:
  - Images, tables, and diagrams as appropriate
  - Referencing citations in APA 7th Edition Referencing style
- Each student group must make a digital copy submission of the following on Canvas.

Use of GenAI

All uses of GenAI in support of your work and in-class discussions must be properly declared, including the specific GenAI used and the prompt used.

Utilizing GenAI to supplement your learning is encouraged; however, any uses of GenAI that would complete your assignments without your critical input or hinder your learning in this class are not encouraged.

If you are in doubt, please consult the tutors.

Class Participation

Students are required to attend all lectures and tutorials. To get the most out of this course, please keep the following expectations in mind:

Schedule & Presence: We meet weekly on Thursday mornings from 9:30 AM to 12:30 PM. Fully engaging for the complete session ensures you stay connected with discussions and group work.

Absences & Exceptions: If you need to miss a class, please reach out in advance whenever possible. Unplanned absences due to illness require a doctor's note to be officially excused. Accumulating three (3) unexcused absences may jeopardize your ability to pass the course.

Impact on Assessment: Attendance and active involvement contribute to your assessment. Meaningfully contributing to group tasks, posing thoughtful questions, and offering constructive feedback during peer presentations provides direct evidence of your critical thinking and engagement.

Class Participation

Class Participation, Self & Peer Assessment (individual)	25%
In-class Individual Written Assessment (individual)	20%
Presentation (group)	25%
Research Report (group)	30%

Schedule

Week	Date & Place	Topic	Milestones
1	03/09/2026 TU201	**[Lecture]** Introduction to Design Research **[Special Lecture]** Prof. Jaden Park **[Tutorial]**	- Group arrangement after lecture
2	10/09/2026 V322	**[Library Workshop]** **[Lecture]** What & Why: Identifying and defining research dimensions; Research Through Design **[Tutorial]**	- Initial project idea / topic - Research purpose - Begin background & precedent review
3	17/09/2026 TU201	**[Lecture]** How #1: Approaches of Qualitative Research & Quantitative Research **[Tutorial]**	- Continue background review - Draft research question / creative proposition
4	24/09/2026 TU201	**[Lecture]** How #2: Data gathering mechanisms + AI **[Tutorial]**	-Submit **Research study project presentation 1** + Research topic + Literature mapping + Inquisitive direction + Research modalities in consideration
-	01/10/2026	***No Class***	- Initiate data gathering process

Schedule

5	08/10/2026	**[Sub-cohort Meeting]** + Design of research modality [workflow] + Data / documentation	**Research study project presentation 2**
6	15/10/2026	**[Tutorial]**	- Adjustments to data collection mechanisms - Data collection
7	22/10/2026 V322	**[Lecture]** How #3: Sensemaking through data filtering and analysis – Descriptive & Analytical **[Tutorial]**	- Continue data collection - Begin data cleansing / editorial selection
8	29/10/2026	**[Sub-cohort Meeting]** + Updated design of research modality [workflow] + Data / documentation + Data filtering procedure	**Research study project presentation 3**
9	05/11/2026 V322	**[Lecture]** How #4 & So What: Communicating and developing design proposition **[Tutorial]**	- Begin data synthesis / analysis
10	12/11/2026	**[Sub-cohort Meeting]** + Updated data / documentation + Data filtering procedure + Data synthesis	**Research study project presentation 4**
11	19/11/2026	**[Tutorial]**	- Begin drawing findings
12	26/11/2026	**[Sub-cohort Meeting]** + Updated data / documentation + Data filtering procedure + Data synthesis + Newly gained insight	**Research study project presentation 5** - Begin drafting the final report
13	03/12/2026 TU201	**[In-class Individual Written Assessment]**	- Complete the research study report and submit by 6 December 2026



Meet the Tutors



Dr. Ar. Mohana Das

Architect & Urban Designer

School of Design
The Hong Kong Polytechnic University

mohana.das@connect.polyu.hk
linkedin.com/in/mohana-das-01606858

Scan for publications (Google Scholar)

TED Tutor

ABOUT ME

I've recently completed my doctorate at PolyU's School of Design, following a Master's in Architecture & Urban Design from Politecnico di Milano, Italy and a Bachelor's and Diploma in Architecture from India. My research bridges design, health and social justice, exploring how design and everyday environments support dignity and wellbeing for older adults.

RESEARCH FOCUS

Aging in Place & LTC
Housing, health & urban form

Salutogenic & Biophilic Design
Nature-integrated wellbeing

Participatory Research
Co-production with older adults

Health & Care Equity
Social justice in aging policy

Dementia & Social Robots
Technology & loneliness in LTC

Inclusive & Universal Design
Access-centered design & pedagogy

Co-Instructor · Research & Analysis for Design (SD5008) · MDes



和睿 (Henry)

khenryhe@gmail.com

rui-he.org

Bachelor of Engineering, Computer Science and Technology
Programming, Computer Vision, Image Processing

Master of Science, Computer Science
Computer Graphics, Machine Learning

Senior Game Engineer
Tang Dynasty Warriors
Tang Dynasty Warriors Mobile
Island Pioneers

PhD, Human–Computer Interaction
Human–AI Collaboration
in Creative Storytelling Practice

IBD Tutor

Hi, I'm Das!



Design Practitioner

8+ years in Business & Design

2026 **PhD |** Design Economics, PolyU, Hong Kong

2019 **MDes|** ID&BM, PolyU, Hong Kong

2016 **MBA |** IIM Shillong, India

Career Roles (4+ yrs) | Sales & Marketing;

Branding - Entertainment Industry & Startup

Publication | 8+ Papers

Startup Funding | Lean Launchpad Program, Alibaba, Micro Fund, HKTechathon, HKSTP

- Co-Design Workshop
- Culture Value Mapping
- Brand Strategy & Identity

- Behavioural Analytics
- AI-Integrated Practice
- Inter-disciplinary Pedagogy

Reach out at dr-bharati.das@connect.polyu.hk

SSD Tutor



Dr. Jennifer Yoohyun Lee

You can call me **Jen**.

I am a practitioner-researcher mostly engaging in the notions of *positioning*, *communication*, *community,* and *infrastructuring*.

jenn-y.lee@polyu.edu.hk

Education Background

- PhD in Social Design (SD, PolyU)
- MA in Design and Environment (Goldsmiths, University of London)
- BFA in Fine Arts (Pratt Institute)

Practice

- Socially Engaged Projects (15+ years | South Korea; USA; UK; Thailand; Hong Kong)
- Curation (4+ years | Hong Kong)

Research Interests

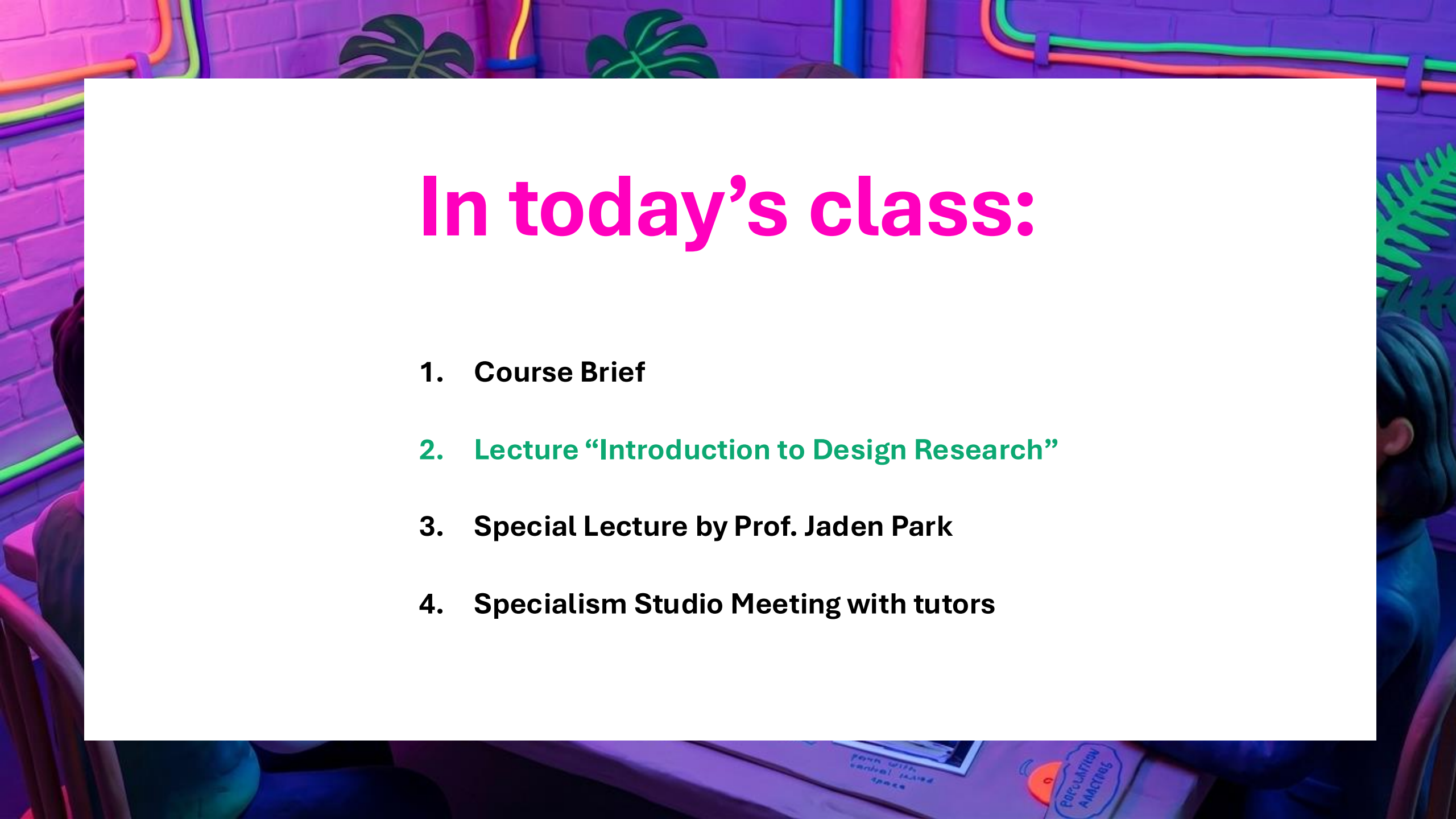
Narrative as talk-in-interaction; Small stories; Discourse analysis; Social media analysis

Design as group decision-making; Participation; Agency; Interpersonal dynamics

Pluriversal communication infrastructuring; More-than-human design

Publication Scope

Design and Narrative; Participatory Design; Social Media Analysis; Urban Nature; More-than-human; Organizational studies; Qualitative; Quantitative; Metanalysis; Mixed Methods +



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Introduction to Design Research

What is “Design”?



Design ← Designare [Latin]

de - + *signare*

“out” + *“to mark”*

= to mark out; to define

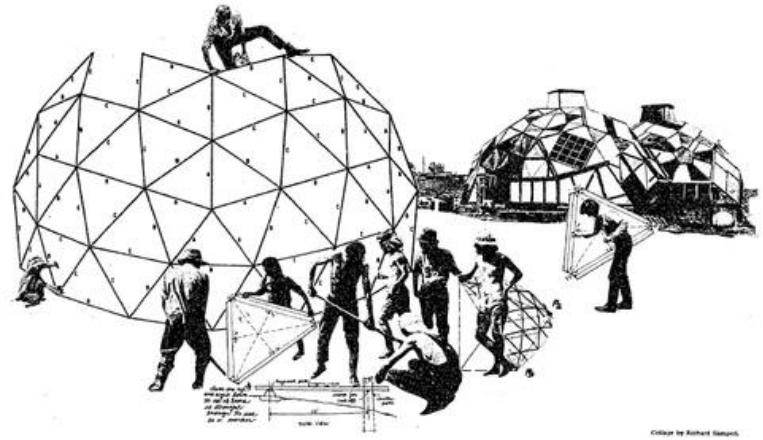
= mapping out an intention, making a mental concept visible

→ <u>composition of aesthetic properties and functional properties for everyday use</u>

What is “Design”?

Design is the ***purposeful transformation*** of an existing state ***into a preferred one***.

- Herbert Simon, *The Sciences of the Artificial* (1969, MIT Press)





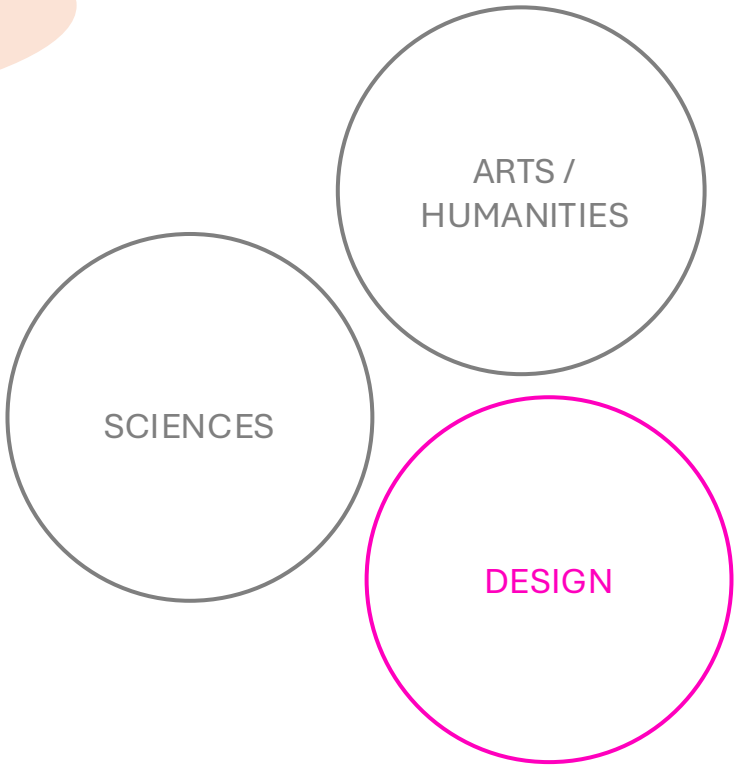


“Drop City” Colorado, USA, 1960 -1979
human-made artifacts and housing not as fixed natural givens, but as practical problem-solving experiments designed to adapt an inner purpose to an outer environment

What is “Design”?

Design is a ***unique cognitive style*** and cultural discipline rooted in ***tacit knowledge, modeling, and practical problem-solving***.

- Nigel Cross, “*Designerly Ways of Knowing*” (1982, Design Studies)



- Natural world
- Controlled experiment, classification, analysis
- Objectivity, rationality, neutrality, and a concern for ‘truth’
- *How things are*

- Human experience
- Analogy, metaphor, evaluation
- Subjectivity, imagination, commitment, and a concern for ‘justice’

- Artificial world
- Modelling, pattern-formation, synthesis
- Practicality, ingenuity, empathy, and a concern for ‘appropriateness’
- *How things ought to be*

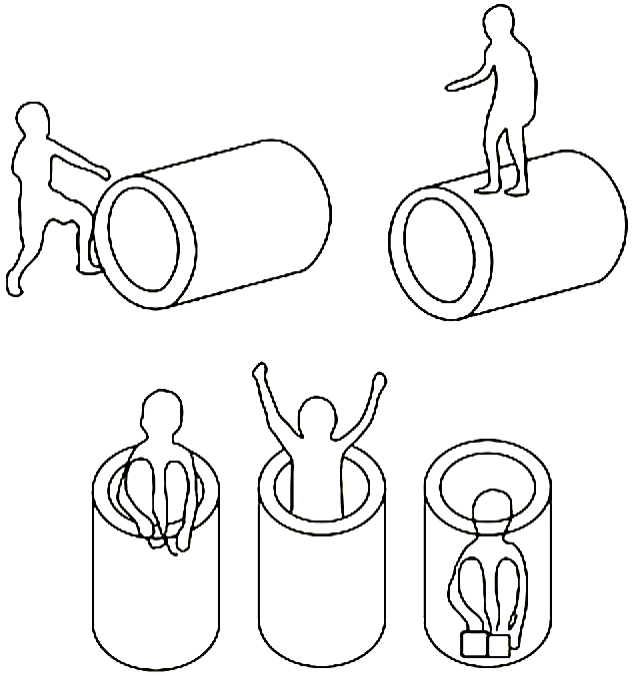
Cross, Nigel. "Designerly ways of knowing." *Design studies* 3, no. 4 (1982): 221-227.

What is “Design”?

Design is fundamentally an act of communication that requires ***deep empathy for how people behave*** rather than how we wish they would behave.

- Don Norman, *The Design of Everyday Things* (1988, MIT Press)





Design as intentional artifacts with affordances - indicating how they could *possibly* be used (Norman, 1988)

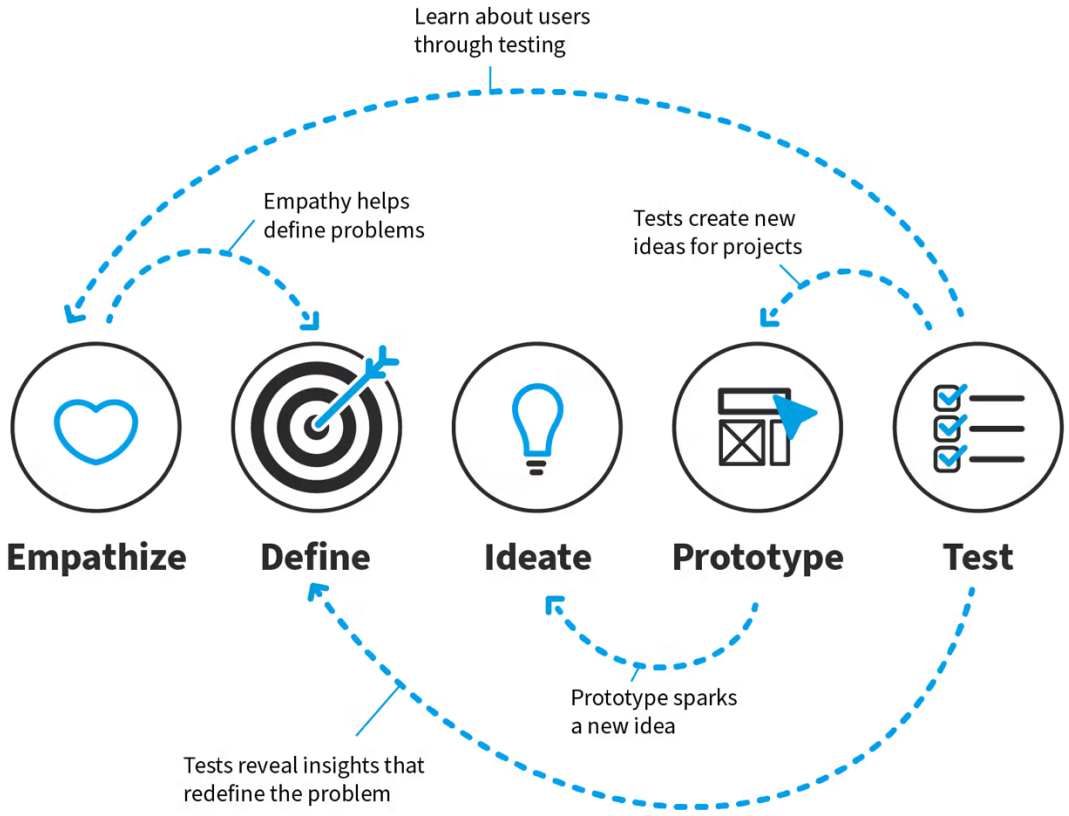
➢ determined by the intentions of designers and usability

Atmodiwirjo, P. (2014). Space affordances, adaptive responses and sensory integration by autistic children. *International Journal of Design, 8*(3).

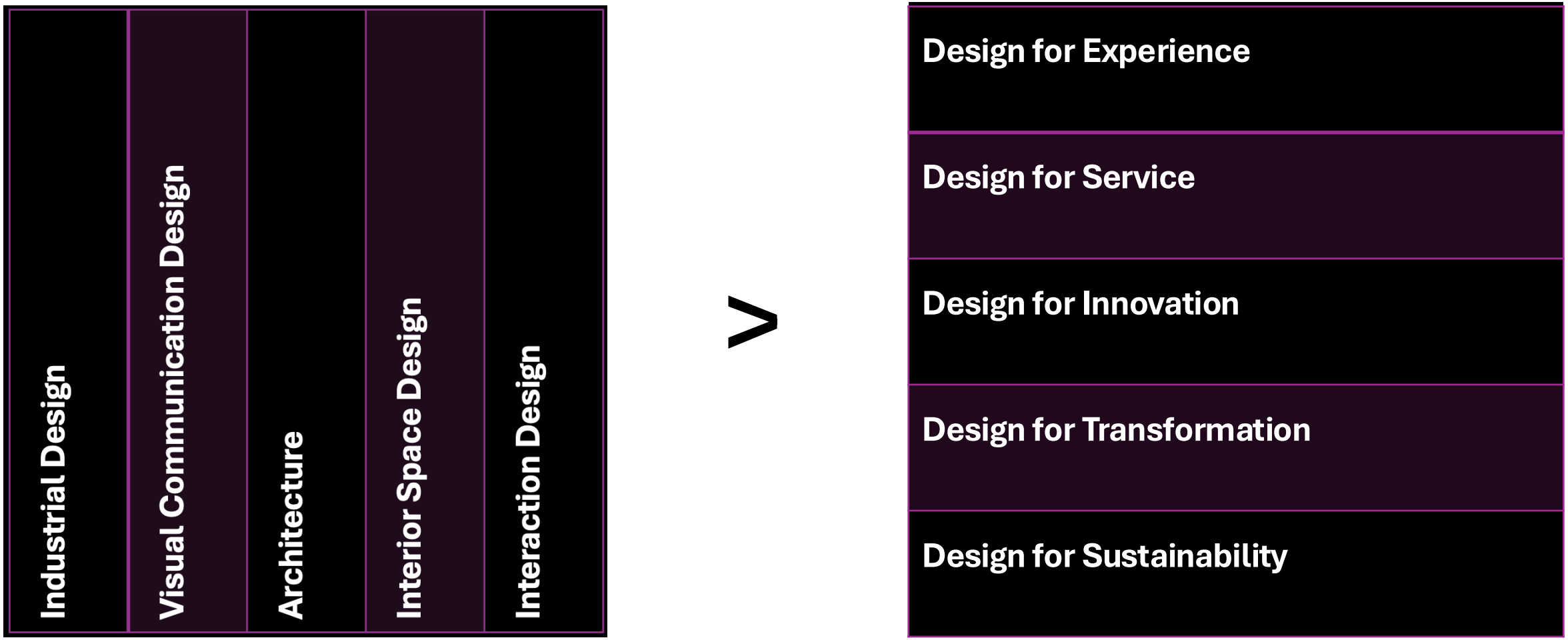
What is “Design”?

Design ***thinking*** is a human-centered framework that balances ***what people desire, what is technologically feasible, and what is economically viable***.

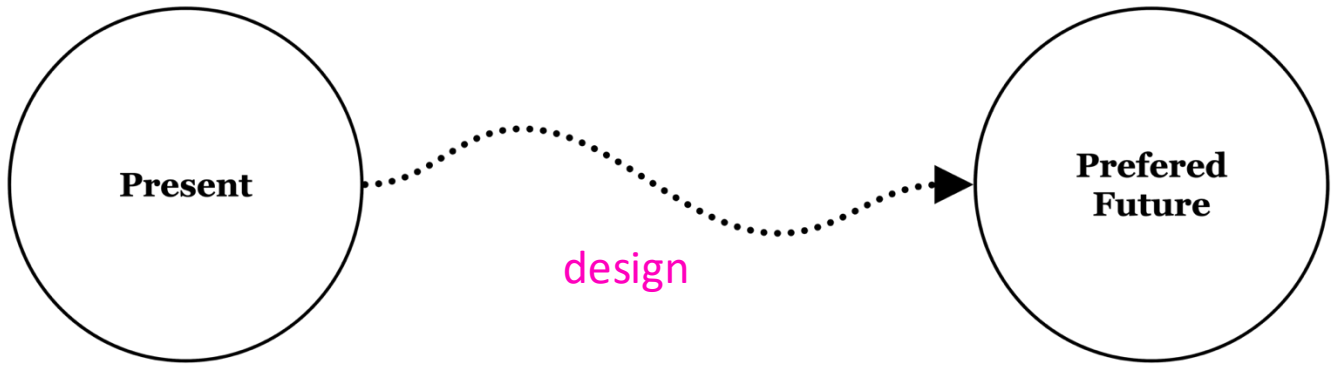
- Tim Brown, *Change by Design: How Design Thinking Transforms Organizations and Inspires Innovation* (2009, HarperBusiness)



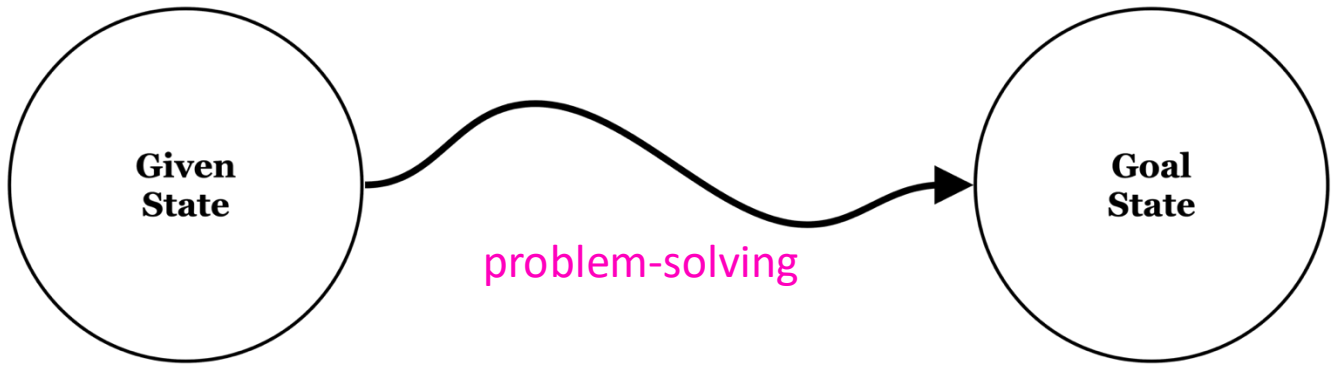
Transformation of “Design”



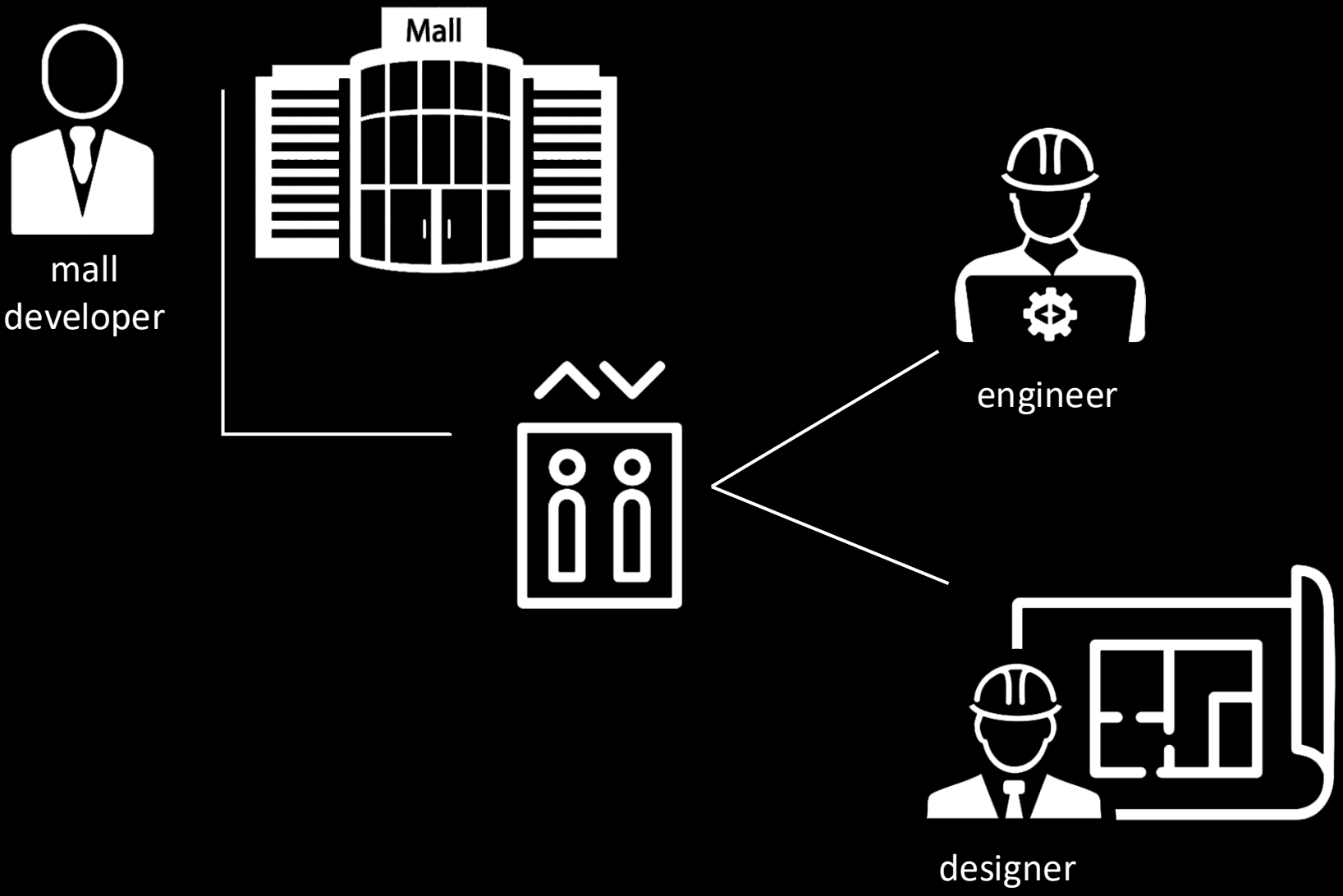
Design as Problem Solving



"everyone designs who devises courses of action aimed at changing existing situations into preferred ones" (Simon, 1969)



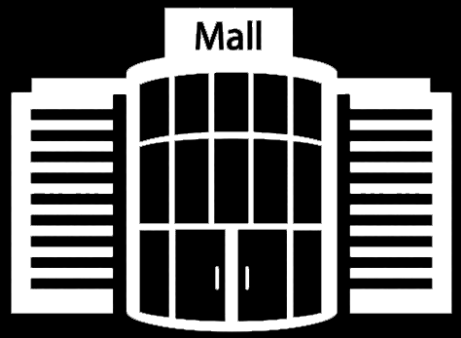
Design as Re-framing Problems



Design as Re-framing Problems



mall developer



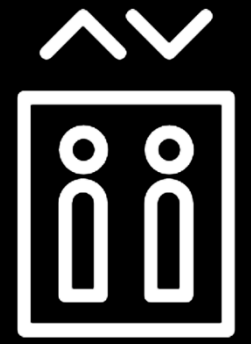
How a problem is framed greatly informs the pathway to a solution



engineer

okay

well-defined problem





designer

?

ill-defined problem



Design as Re-framing Problems



https://www.dreamstime.com/photos-images/elevator-mirror.html

Design Context

- The increasing complexity of contemporary situations and technologies is driving rapid growth in the connections among science, engineering, psychology, and design.
- Because of this, the relationship between research and design is becoming tighter.



What is “Research”?

“**Research**”, generally speaking, is:

- a careful and *organized way of asking questions, gathering information, and studying data*.
- an activity to *find new facts, solve problems, or learn more about a topic*.
- an endeavor to move past guessing to build true understanding through a clear plan.



https://www.facebook.com/groups/948169185609831/posts/2363906894036046/

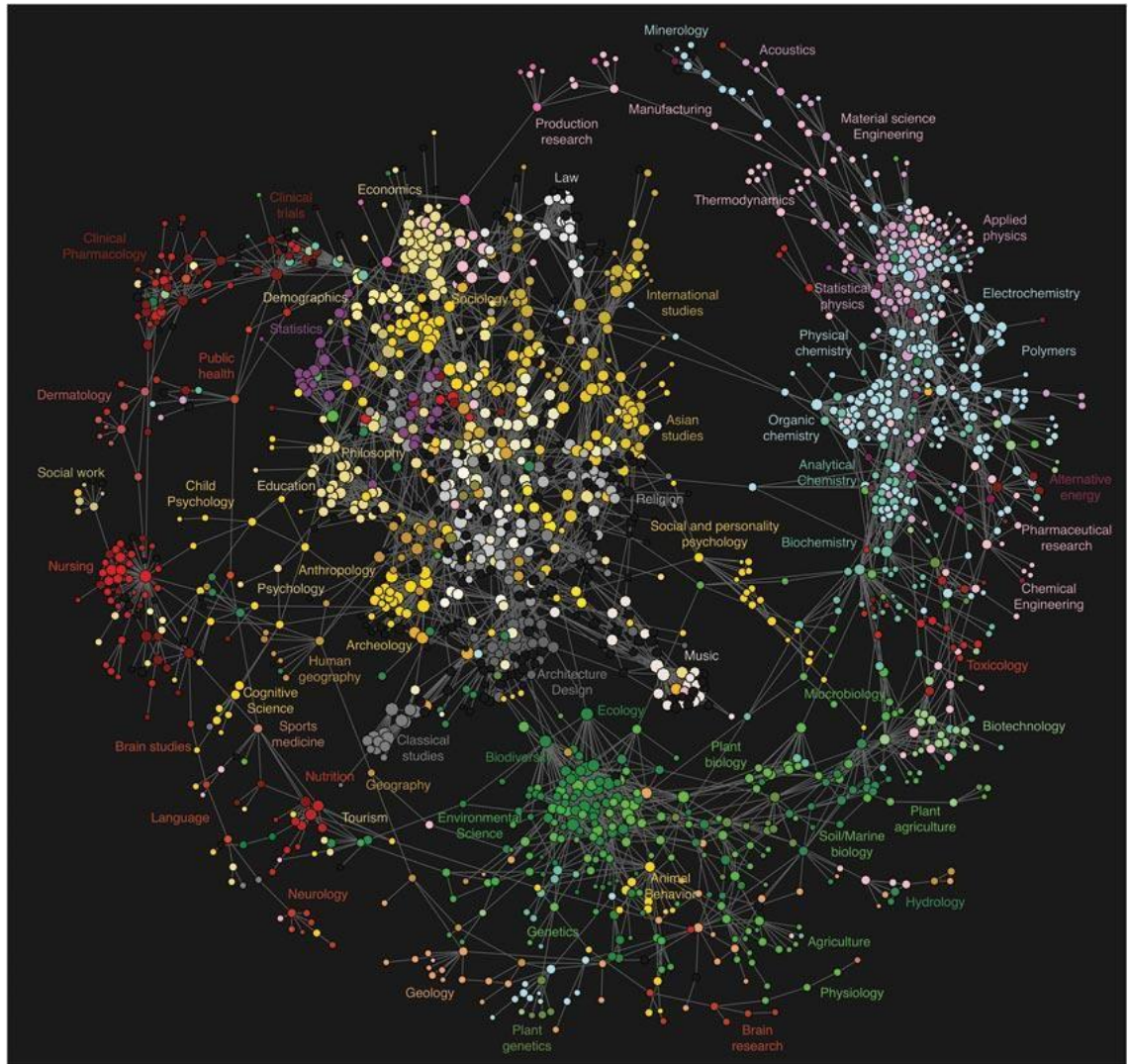
Types of Research

Basic Research	Explores a topic to gain deep knowledge without immediate practical use.	It is purely to gain knowledge about the world, with no immediate practical application or product in mind. e.g., studies on human DNA, studies on stories of the last generation of traditional deep-sea fishermen, etc.
Applied Research	Focuses on fixing a specific, practical problem.	It directly focuses on solving a specific, practical, real-world problem. e.g., studies on improving fiber optics, studies on treating depression, etc.

Purposes of Research

Academic Research

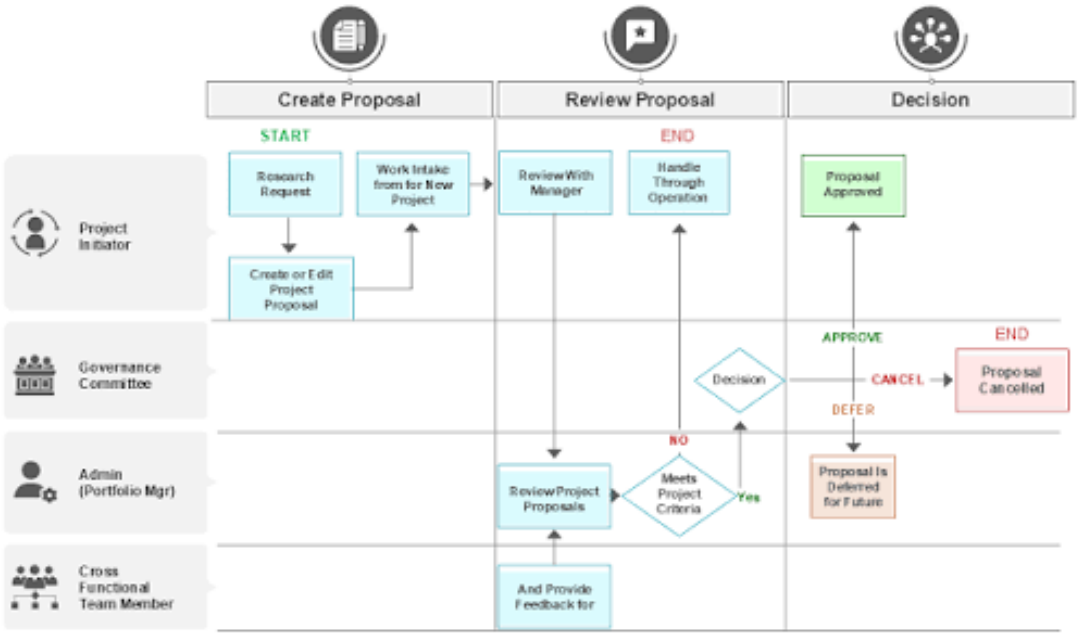
- Researchers / scholars follow strict rules, check previous work, test their ideas using real data, instead of personal opinions or gut feelings.
- Other experts must review their work before it is shared with the world.
- Its goal is *to add new insights to the pool of knowledge*.



Purposes of Research

Administrative Research

- Industries study how they can operate better by finding bottlenecks, cutting waste, improving management, and ensuring compliance.
- Administrative research values efficiency, compliance, accuracy, and support.

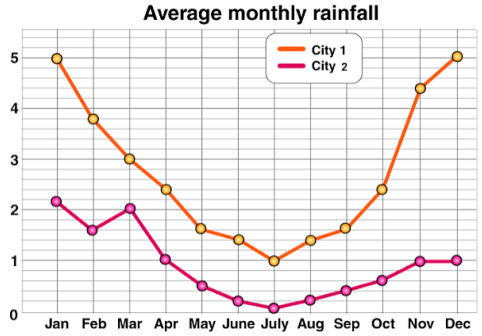


https://www.slideteam.net/blog/top-10-project-workflow-templates-with-samples-and-examples

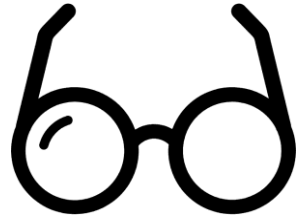
Purposes of Research – Choose your direction

Academic Research	**Primary Objective**	Produce novel *knowledge*, theoretical models, and methodological frameworks.
	Validation Standard	Methodological rigor, *peer-reviewed validation, replicability, and theoretical soundness*.
	Primary Audience	Academic peers, research communities, scholarly journals, and conferences.
	Output Formats	Peer-reviewed papers, academic monographs, validated design frameworks, and doctoral theses.
Administrative Research	**Primary Objective**	Provide *actionable insights* to resolve specific organizational, market, or product challenges.
	Validation Standard	*Methodological soundness*, strategic utility, commercial viability, risk reduction, speed, and *stakeholder consensus*.
	Primary Audience	Corporate executives, product managers, design teams, clients, and investors.
	Output Formats	Executive briefs, strategic roadmaps, design requirement documents, journey maps, and commercial prototypes.

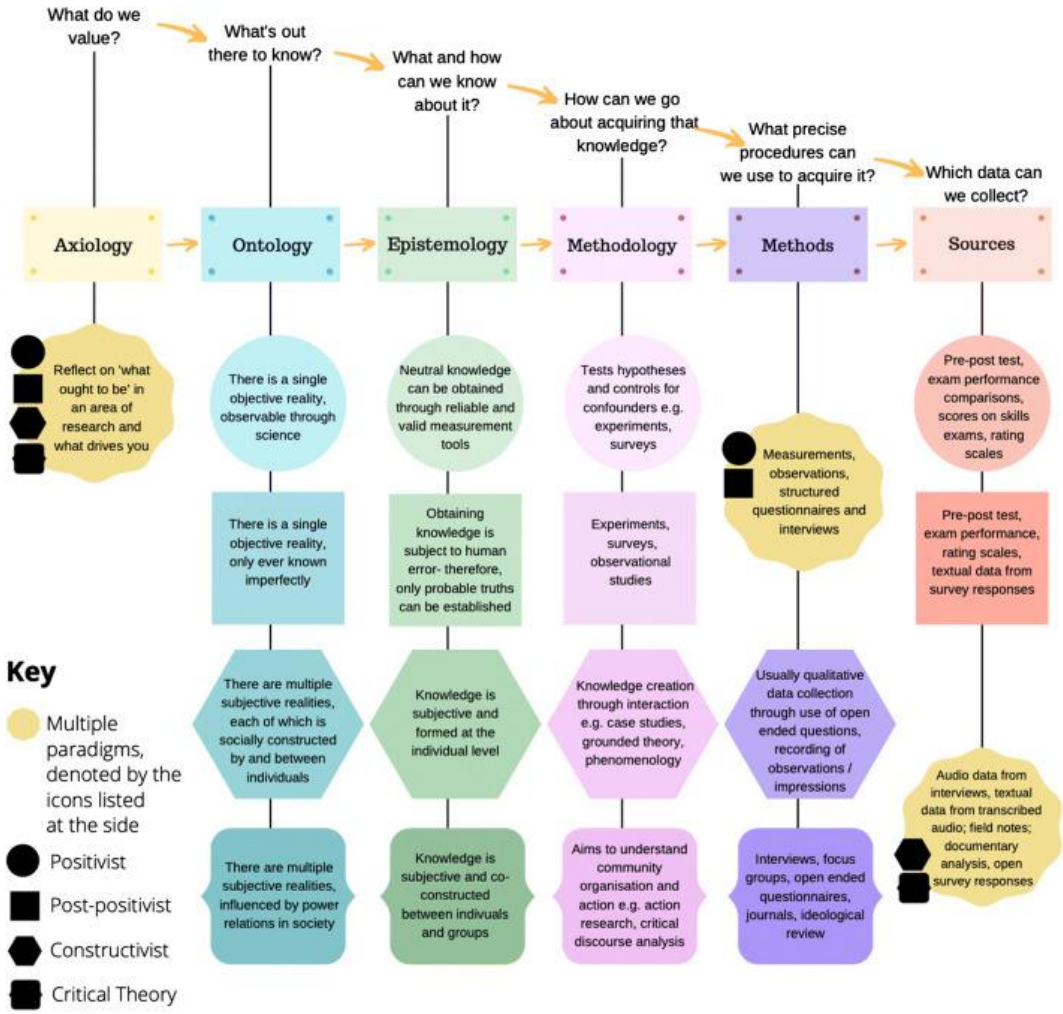
Types of Research - Data

Qualitative Research	Gathers words, opinions, and experiences through interviews or chats to understand thoughts and reasons.	It is appropriate for gathering deep personal feelings, opinions, and experiences instead of statistics. 
Quantitative Research	Gathers numbers and measurements using surveys, tests, or math to track trends and test ideas.	It relies entirely on hard numbers, percentages, and math to find patterns. 

Research Paradigms

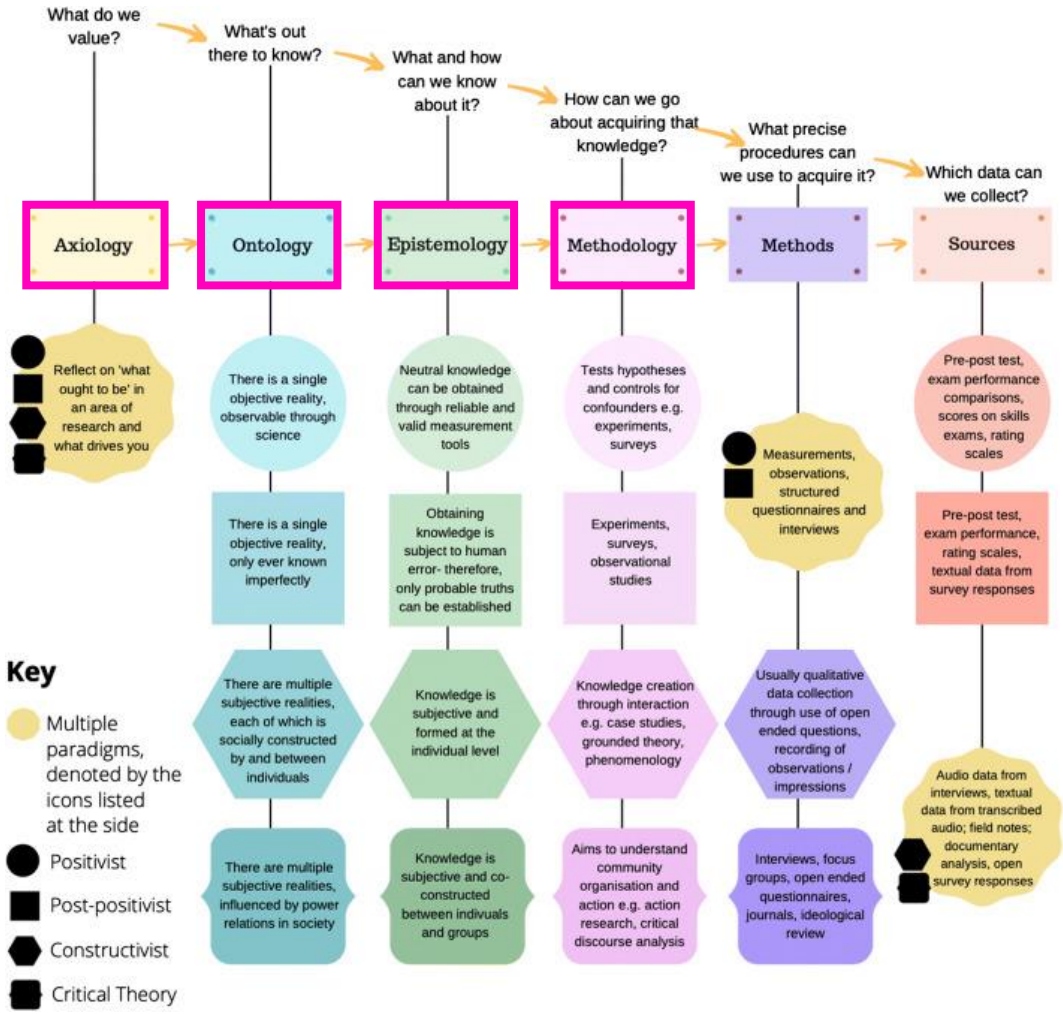


- A research paradigm is the conceptual “pair of glasses” one wears when looking at the world
- They are foundational beliefs that determine what you notice, how you define reality, and how you interpret what you see.
- Research paradigms evolved over centuries of philosophical inquiries as thinkers realized that studying different things requires fundamentally different assumptions.



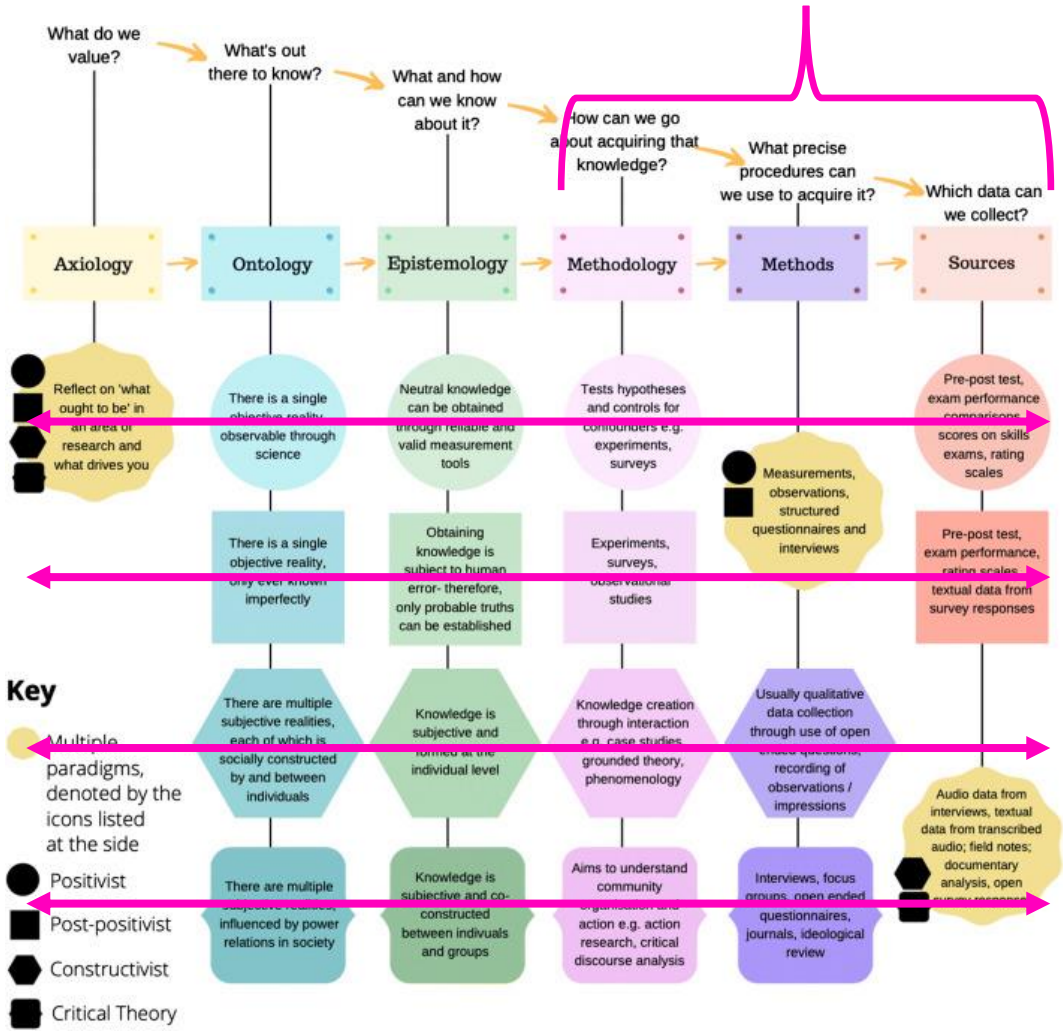
Research Paradigms

- Ontology, Epistemology, Axiology, and Methodology are the four structural dimensions of research philosophy.
- **Ontology (The Nature of Reality):** What exists?
- **Epistemology (The Nature of Knowledge):** How do we know what we know?
- **Axiology (The Role of Values & Ethics):** What is valuable?
- **Methodology (The Strategy of Inquiry):** How do we investigate?

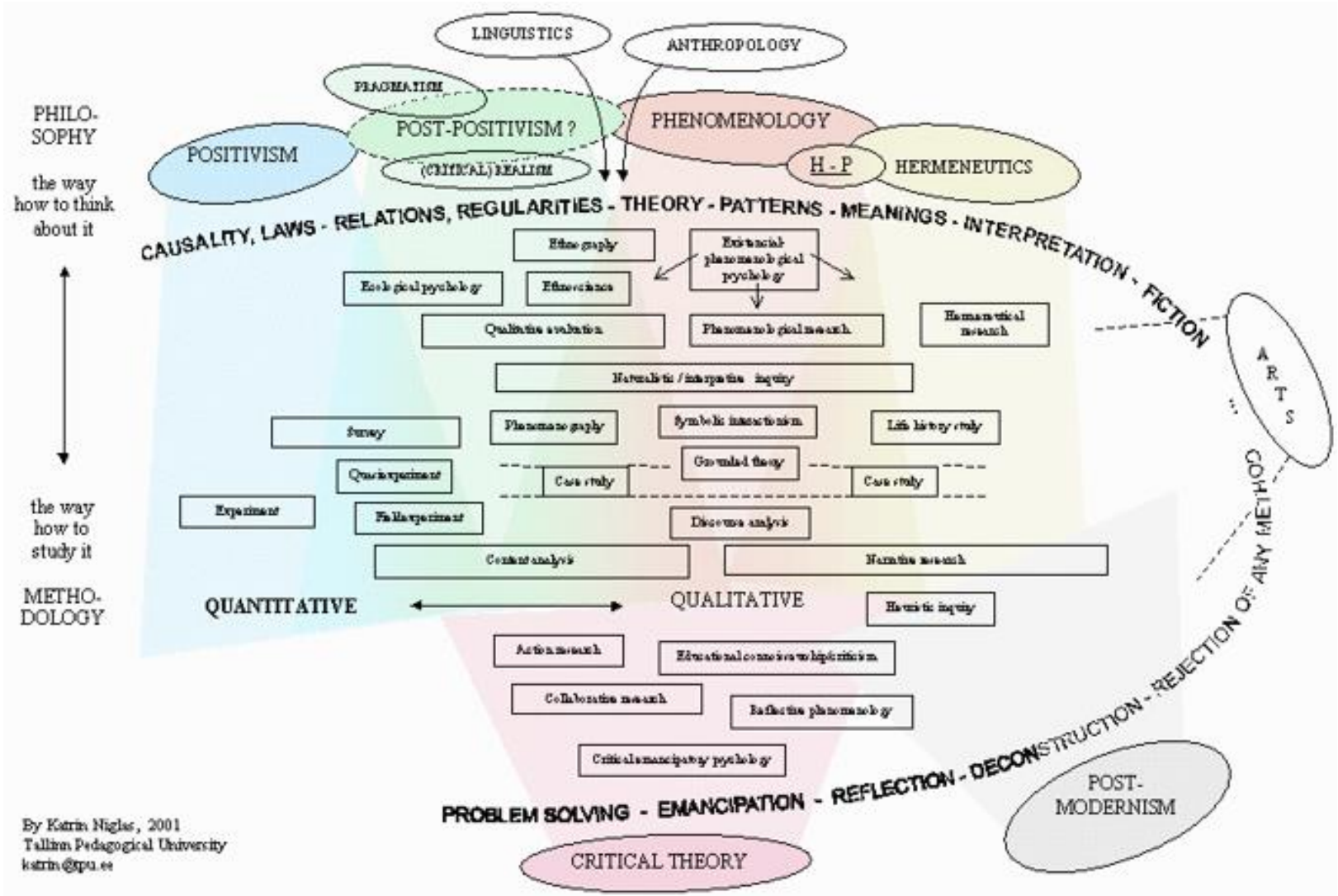


Research Paradigms

- A research paradigm is created when a researcher takes a specific stance across all four pillars.
- The specific paradigms (the complete worldviews) are formed by answering those questions.



Research Paradigms



Research Paradigms

Positivism

- **Ontology:** A single, objective reality exists independently of human perception, governed by universal natural laws.
- **Epistemology:** Knowledge is discovered through detached observation; the researcher and subject remain separate so facts can be measured without bias.
- **Axiology:** Research must be strictly value-free, eliminating personal values and emotions.
- **Methodology:** Quantitative experiments, controlled measurements, statistical modeling, and formal hypothesis testing to establish cause-and-effect relationships.

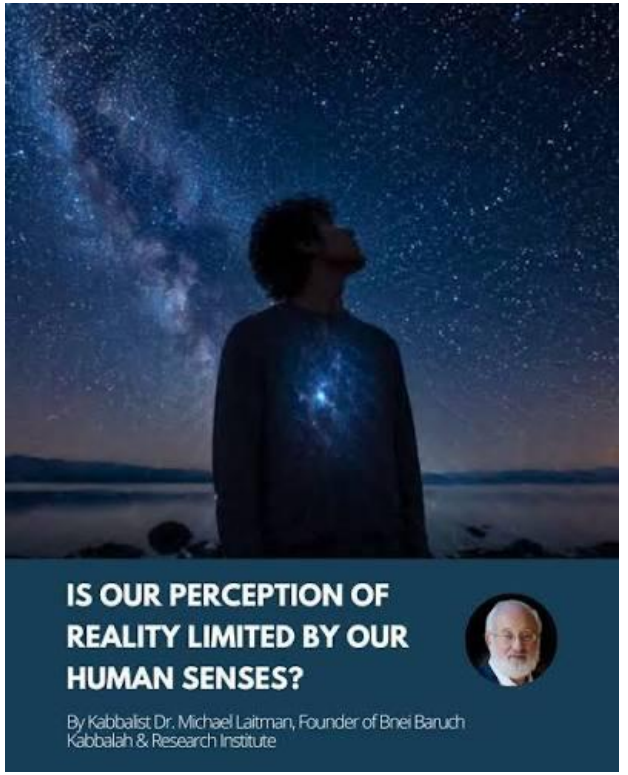


https://sociologytwynham.com/2014/12/17/positivism-the-scientific-approach-to-social-research/

Research Paradigms

Post-positivism

- **Ontology:** An objective reality exists, but human perception is inherently fallible, so reality can only be known imperfectly and probabilistically.
- **Epistemology:** Absolute objectivity is unachievable, but researchers strive toward it through peer critique, falsification, and triangulation.
- **Axiology:** Researcher biases can influence findings, so systematic methods are needed to identify, control, and minimize bias.
- **Methodology:** Deductive experimentation, quantitative measures, and multi-method triangulation aim to falsify hypotheses rather than prove absolute truth.



https://www.instagram.com/p/DZKqIsEEilm/

Research Paradigms

Pragmatism

- **Ontology:** Reality is dynamic, contextual, and shaped by actions, experiences, and consequences, not a static, independent truth.
- **Epistemology:** Knowledge is a tool for action; truth is measured by its practical utility, problem-solving effectiveness, and "what works" in a given situation.
- **Axiology:** Inquiry is guided by human values and ethics, focusing on real-world impact, consequences, and desirability.
- **Methodology:** Inquiry is problem-driven, freely combining qualitative, quantitative, and design tools to address challenges.



https://www.facebook.com/Ranker/posts/sure-people-will-probably-understand-you-if-you-use-the-term-jerry-rigged-but-th/1456821803146500/

Research Paradigms

Phenomenology

- **Ontology:** Reality includes conscious, subjective experiences directly perceived and lived, not an objective external world.
- **Epistemology:** Knowledge is gained by uncovering the core essence and personal meaning within first-person human experiences.
- **Axiology:** Researchers acknowledge biases and actively set aside assumptions to focus on the participant's authentic voice.
- **Methodology:** Inquiry uses phenomenological interviews, reflective journaling, thematic analysis, and autoethnography to capture detailed lived accounts.

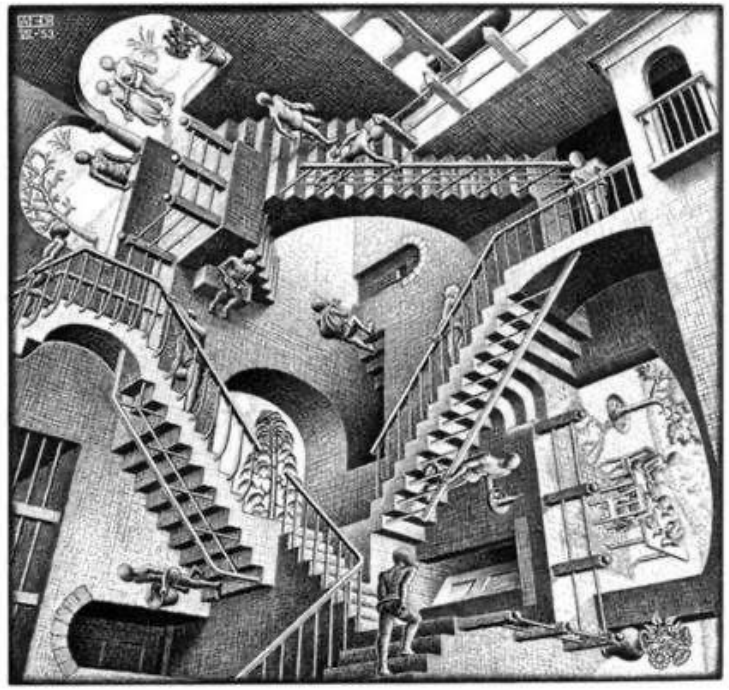


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Research Paradigms

Interpretivism

- **Ontology:** Reality is subjective, multiple, and socially constructed through individual experiences, cultural contexts, and social interactions.
- **Epistemology:** Knowledge is co-constructed through empathetic interaction, interpretation, and dialogue between researcher and participants.
- **Axiology:** Researchers recognize their own background, biases, and positionality as central to interpreting data.
- **Methodology:** Inquiry uses ethnography, interviews, grounded theory, and thematic analysis to grasp the meanings people assign to their experiences.

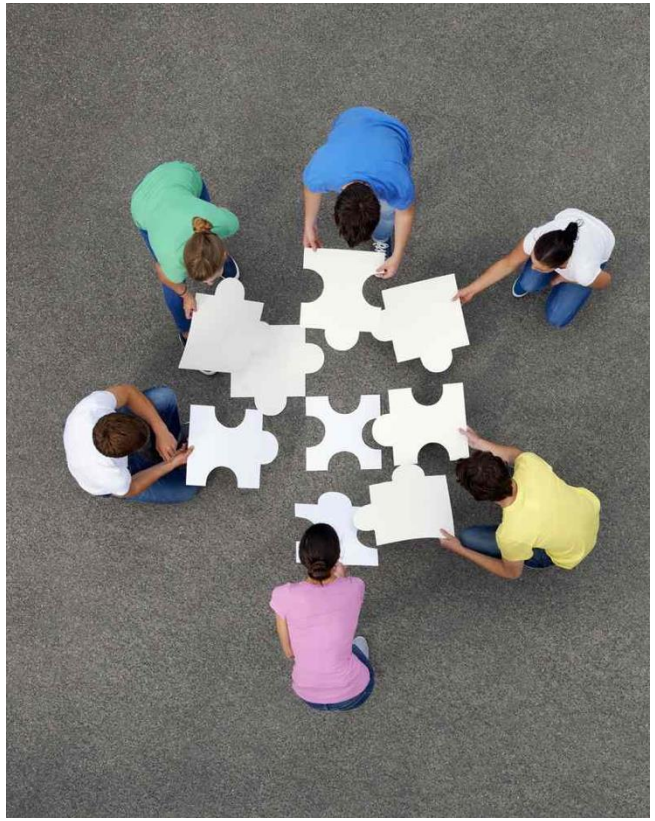


https://research-methodology.net/research-philosophy/interpretivism/

Research Paradigms

Social constructivism

- **Ontology:** Reality is continually constructed, negotiated, and maintained through human interaction, language, and cultural practices.
- **Epistemology:** Knowledge is social and historically situated; truth is co-created through dialogue, interpretation, and relationships, not individual observation.
- **Axiology:** Both researchers and participants view the world through socially conditioned lenses.
- **Methodology:** Inquiry uses collaborative approaches like co-design, participatory research, focus groups, discourse analysis, and interviews to study how communities create and navigate shared meaning.

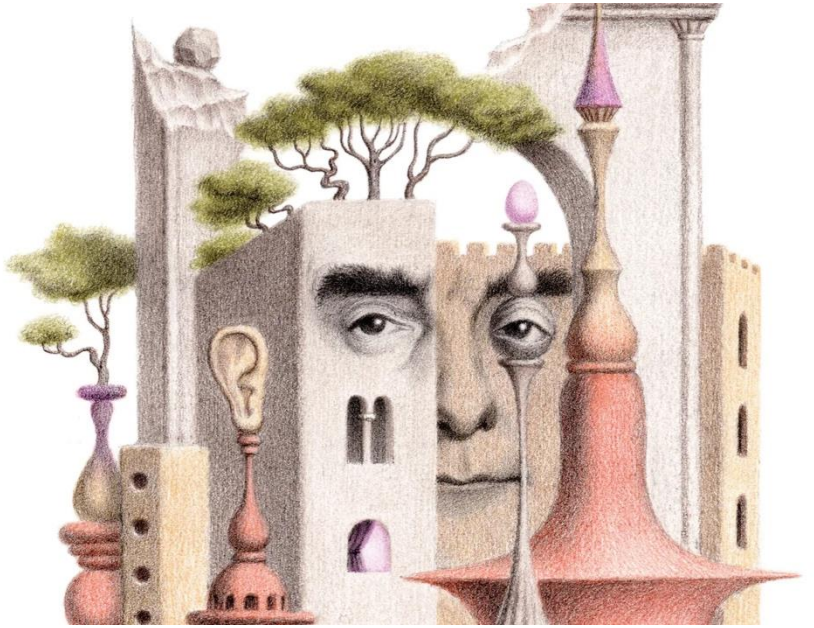


https://www.thoughtco.com/social-constructionism-4586374

Research Paradigms

Postmodernism

- **Ontology:** Reality is fragmented, unstable, and constructed entirely through language, discourse, and power, rejecting universal truths and grand metanarratives.
- **Epistemology:** Knowledge is historically and culturally contingent, bound to power structures.
- **Axiology:** Researchers aim to challenge dominant ideologies, expose hidden power dynamics, and center subverted or marginalized voices.
- **Methodology:** Inquiry uses discourse analysis, visual semiotics, textual deconstruction, autoethnography, and non-linear narrative analysis to challenge fixed meanings and reveal assumptions.



https://www.newyorker.com/tag/postmodernism

Research Paradigms

Critical Theory

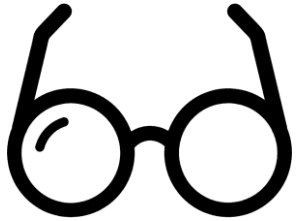
- **Ontology:** Reality is influenced by social, political, economic, and cultural power dynamics that are accepted, often oppressive, as natural structures.
- **Epistemology:** Knowledge is shaped by power; understanding involves revealing the ideological interests behind societal 'truth.'
- **Axiology:** Research is normative and activist, driven by a moral commitment to promote social justice.
- **Methodology:** Inquiry uses critical discourse analysis, participatory action research, historical critique, and consciousness-raising dialogue to empower communities and drive systemic change.



https://www.linkedin.com/pulse/ethical-commitment-organization-where-do-we-begin-james-dillard

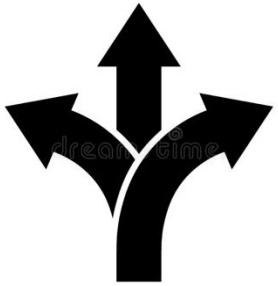
Find out what your general worldview might be:

From Worldview to Thinking Rules



Starting assumptions

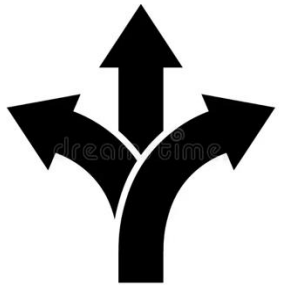




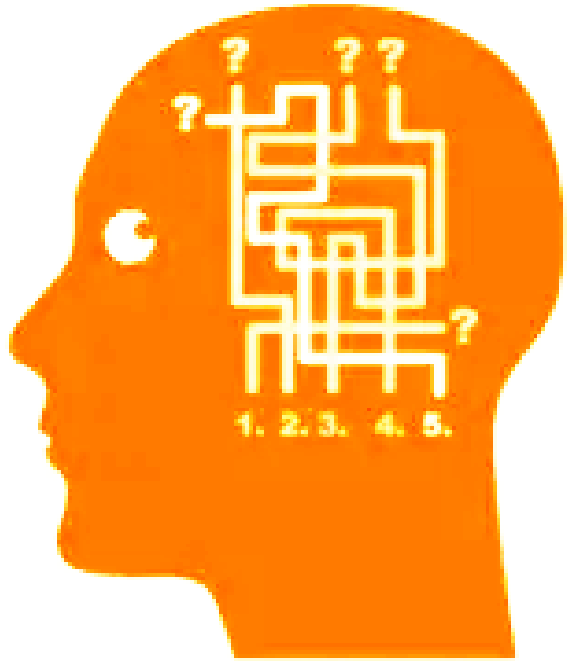
Connecting observations to final conclusions, to be free of false assumptions

- A paradigm is your basic belief about how the world works and what “truth” is.
- Logic is the set of step-by-step thinking rules you use to make sensible, reliable conclusions based on those beliefs.

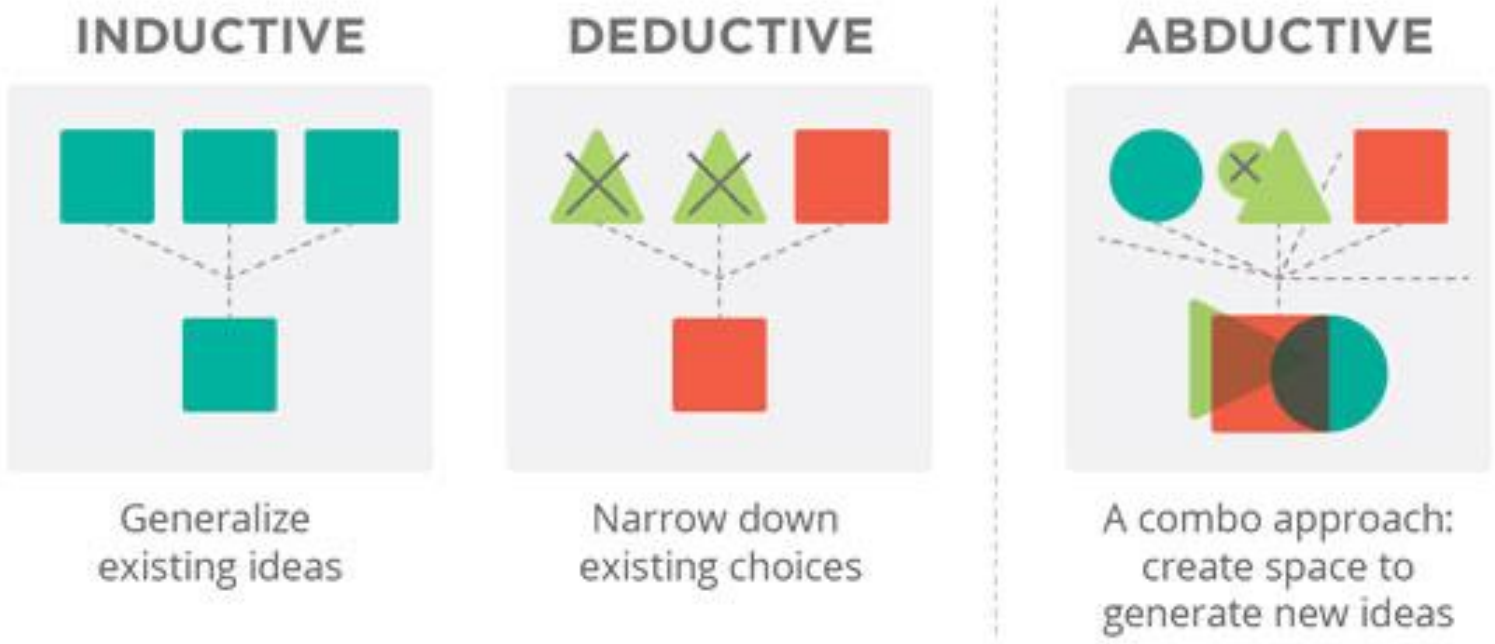
Logic in Research



- Logic and reasoning form the structural backbone of research.
- When information provides the raw material of reality, logic supplies the formal structure, rules, and inferential mechanics needed to process.
- It translates raw empirical data into defensible insights and ensures that every claim is grounded in sound inference.
- Without logical rigor, research devolves into *unexamined assumptions and disconnected anecdotes*.



Types of Logic



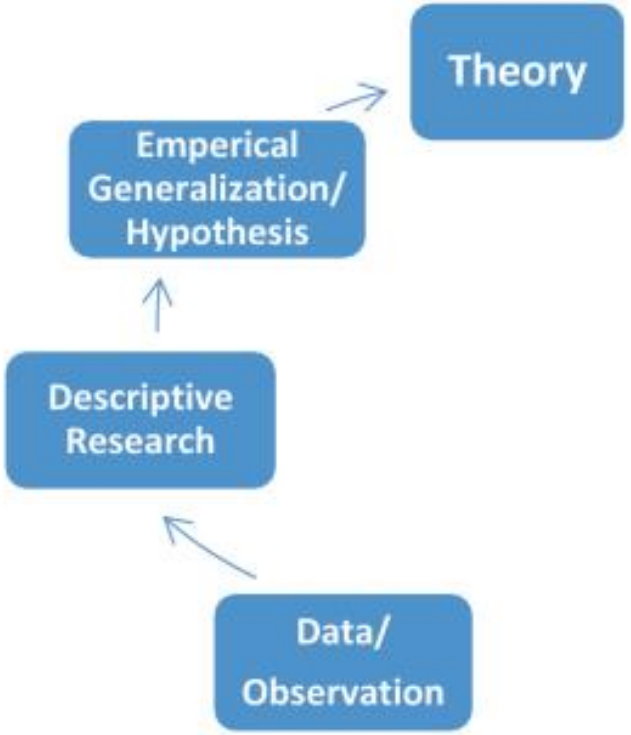
Logic is the **basis of critical thinking** and **how we make sense** of complexity.

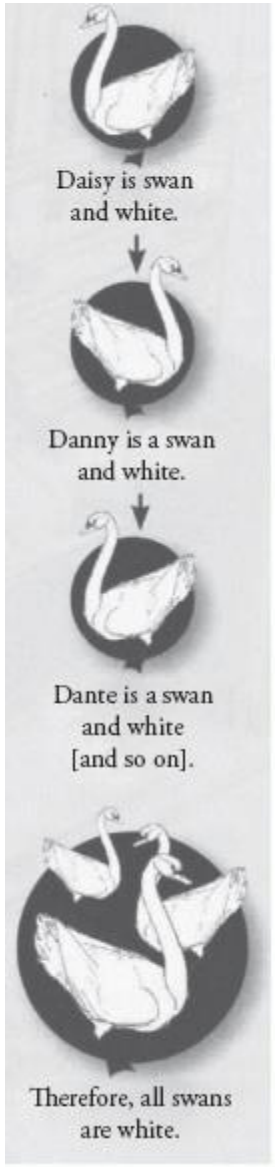
Types of Logic

Inductive

- Reasoning from specific observations to a broader view
- Going from the specific to the general





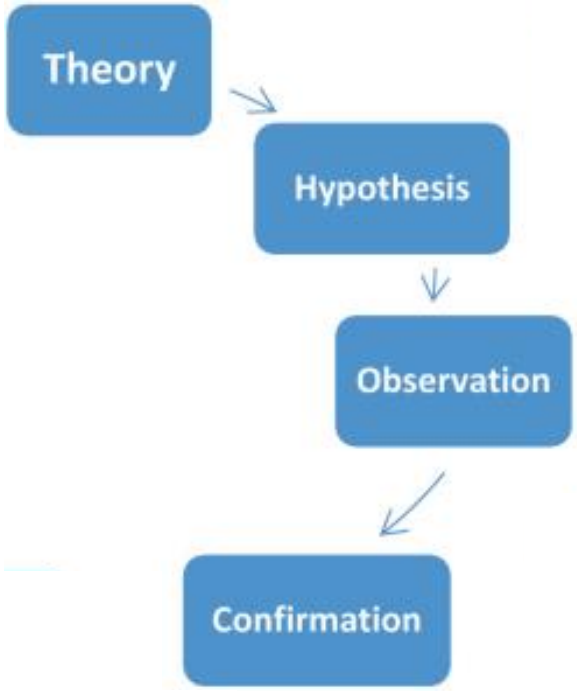


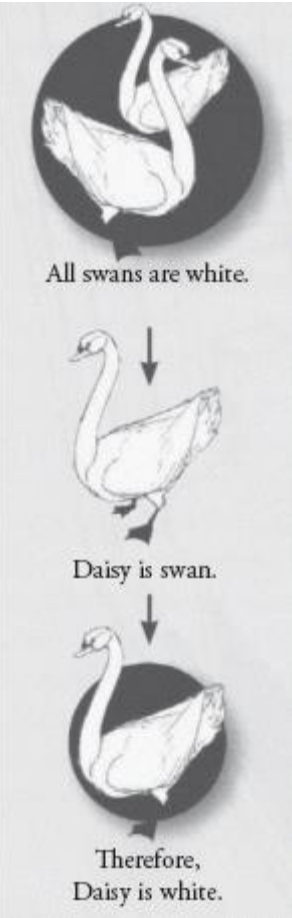
Types of Logic

Deductive

- Parts of the logic combine to make the conclusion true
- Going from the general to the specific



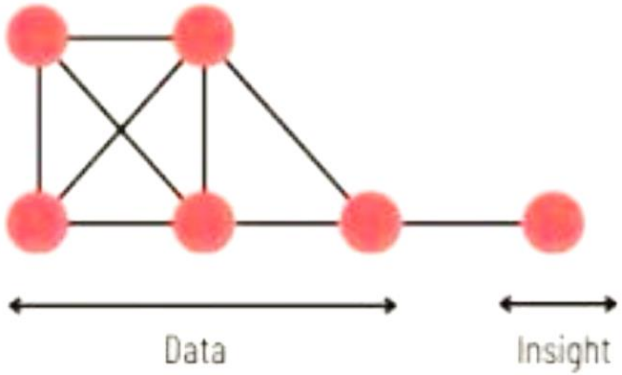


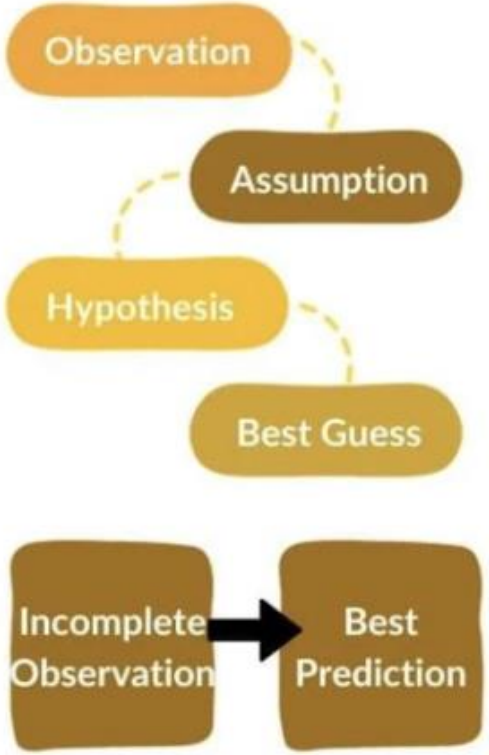


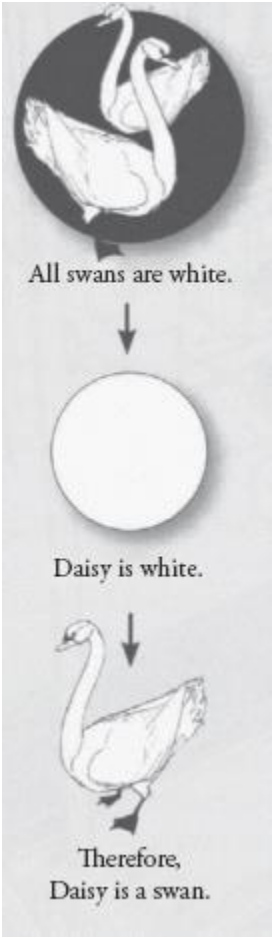
Types of Logic

Abductive

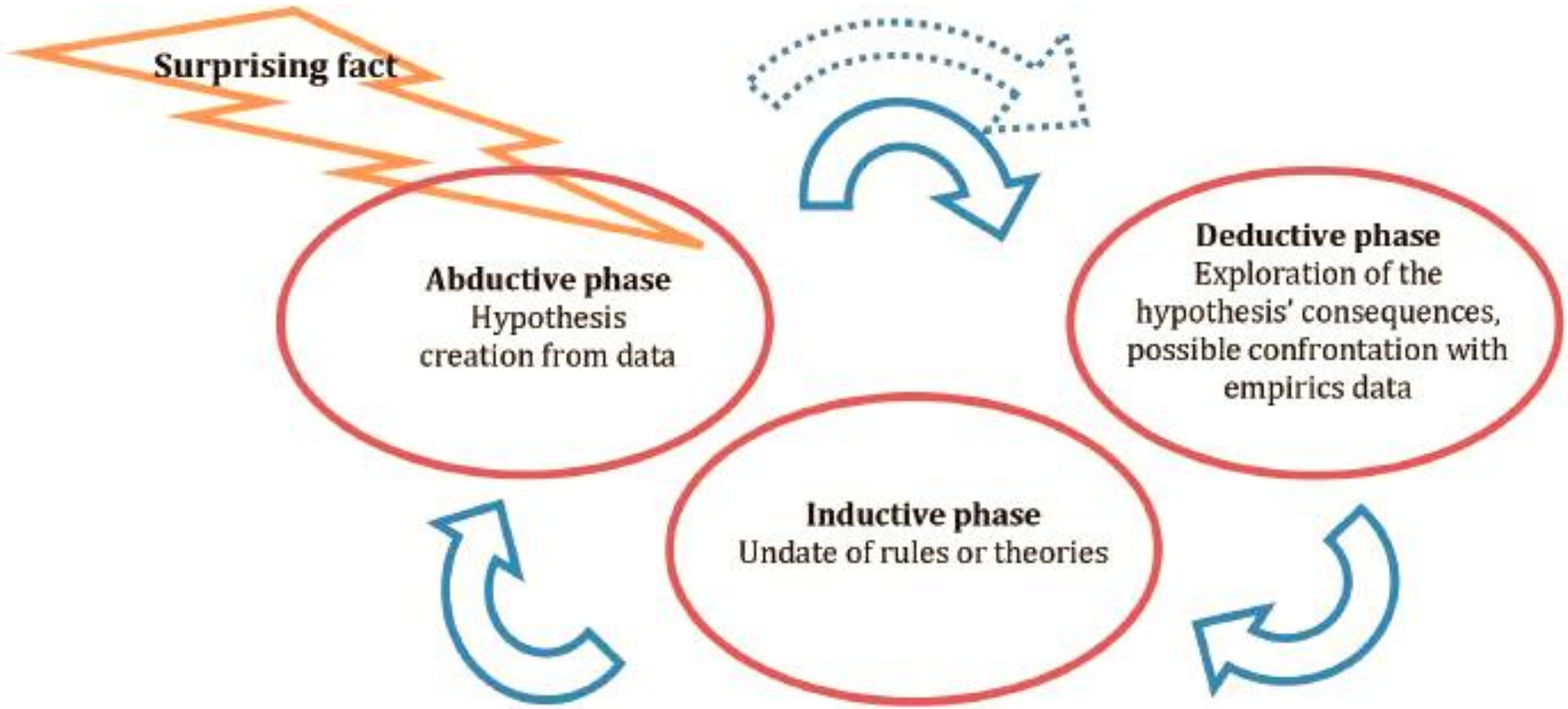
- Looking at input (often from different sources) and connecting the dots to come up with an educated guess
- Finding the best explanation







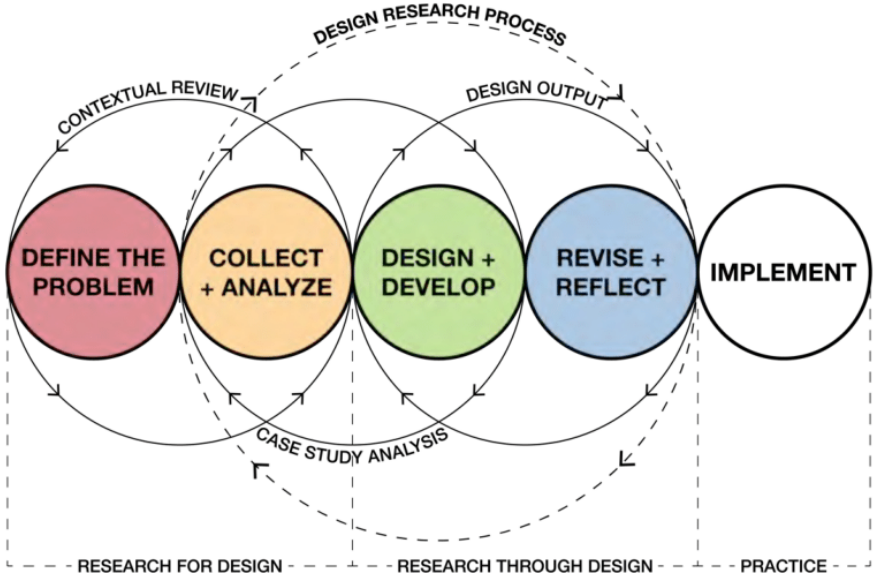
Types of Logic



What is “Research” for designers?

“**Research**” in design context is:

- systematic gathering of information, context, and user needs to shape, test, and validate *creative solutions*.
- *transforms assumptions into evidence-based decisions*, ensuring that user problems and institutional goals are genuinely solved.

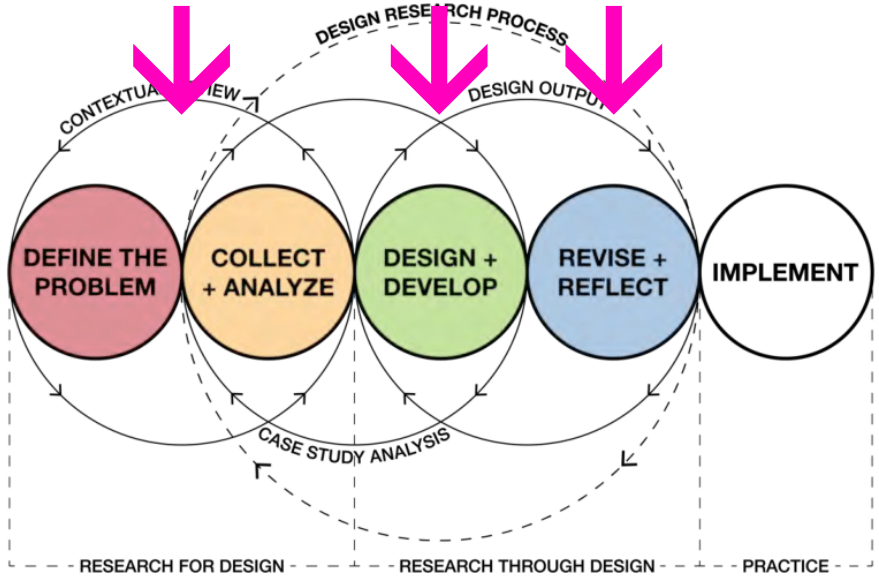


https://www.researchgate.net/figure/A-diagrammatic-representation-of-the-authors-design-research-process_fig1_338421707

What is “Research” for designers?

“**Research**” in design context is for:

- **Generative/Discovery:** Uncovering hidden user needs and behaviors before creation.
- **Evaluative/Testing:** Checking if prototypes or live designs work for users.
- **Market/Contextual:** Analyzing competitors, trends, and technical limits.



https://www.researchgate.net/figure/A-diagrammatic-representation-of-the-authors-design-research-process_fig1_338421707

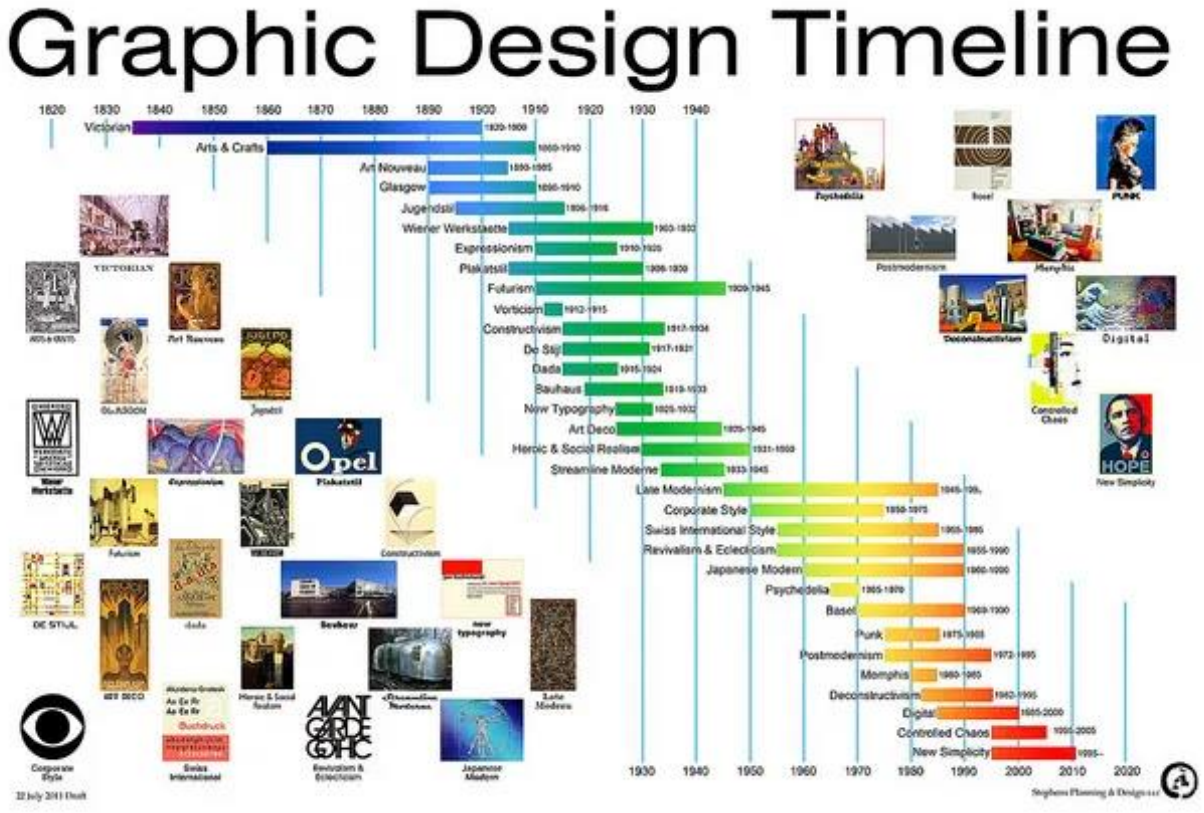
Types of Design Research - Relationship

Research *into* Design	• *Observes the design world from a distance to understand its history, behavior, and cultural impact*. • Analyzes the context, culture, sociology, or aesthetics of design from an external perspective. • Results typically in a traditional written thesis or dissertation.	Designed artifact is Deconstructed / Critiqued
Research *for* Design	• Involves gathering reference materials, user needs, and technical data to *inform a future project*. • Operates as background preparation - such as market research, ergonomics, or material specs - where the final output is the design artifact itself.	Designed artifact is the intended end goal
Research *through* Design	• *Uses the actual process of making and designing as a primary method of inquiry*. • Includes material exploration, digital development work, and action research. • Combines studio-based experimentation with a written report to document tacit knowledge.	Designed artifact is an active testing tool

Types of Design Research - Relationship

Research *into* Design

- Focuses on studying past work, the design process, and how designers work from perspectives like psychology, history, and philosophy.
- **Core focus:** Examining design itself and designers as the primary subjects of study.
- **Goal:** Understanding how design works to build educational materials, write design history, and improve professional tools.



https://graphicdesignthoughts.blog/courses/arth-101-history-of-design-thinking/timelines/

Types of Design Research - Relationship

Research *for* Design

- Often doesn't look like design research because it studies human behavior and systems.
- Focuses on gathering information, user needs, and context before starting a project to guide creative and strategic decisions.
- **Core focus:** Information-gathering (like user interviews, benchmarking, and market research) prior to building an artifact.
- **Goal:** Ensuring building the right solution based on solid evidence rather than guesswork.



https://www.neonscience.org/impact/observatory-blog/calibrating-remote-sensing-data-field-observations-east-river-watershed

Types of Design Research - Relationship

Research *through* Design

- Focuses on using the hands-on process of building prototypes and artifacts to generate new knowledge.
- **Core focus:** Using prototypes as tangible questions or tests rather than final commercial products.
- **Goal:** Designing, testing, reflecting, and redesigning in cycles to uncover insights about user behavior and technology.



https://ixdf.org/literature/book/the-encyclopedia-of-human-computer-interaction-2nd-ed/research-through-design

Design Research Landscape

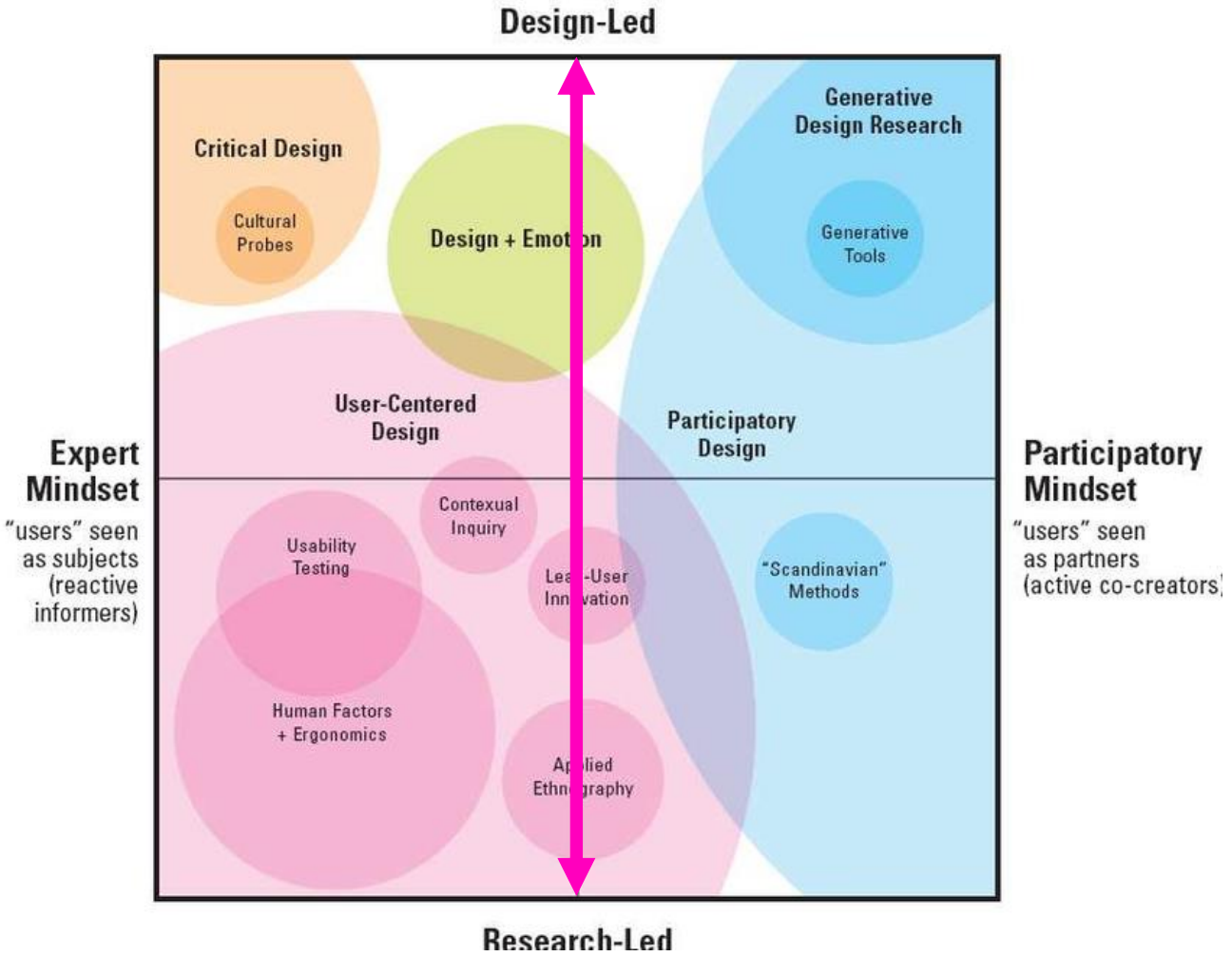
Approaches

Research-led:

- Traditional approach
- Driven by applied psychologists, anthropologists, sociologists, and engineers.

Design-led:

- Recent approach
- Driven by design practitioners



Sanders, E. B. N., & Stappers, P. J. (2012). Convivial toolbox: Generative research for the front end of design. Bis.

Design Research Landscape

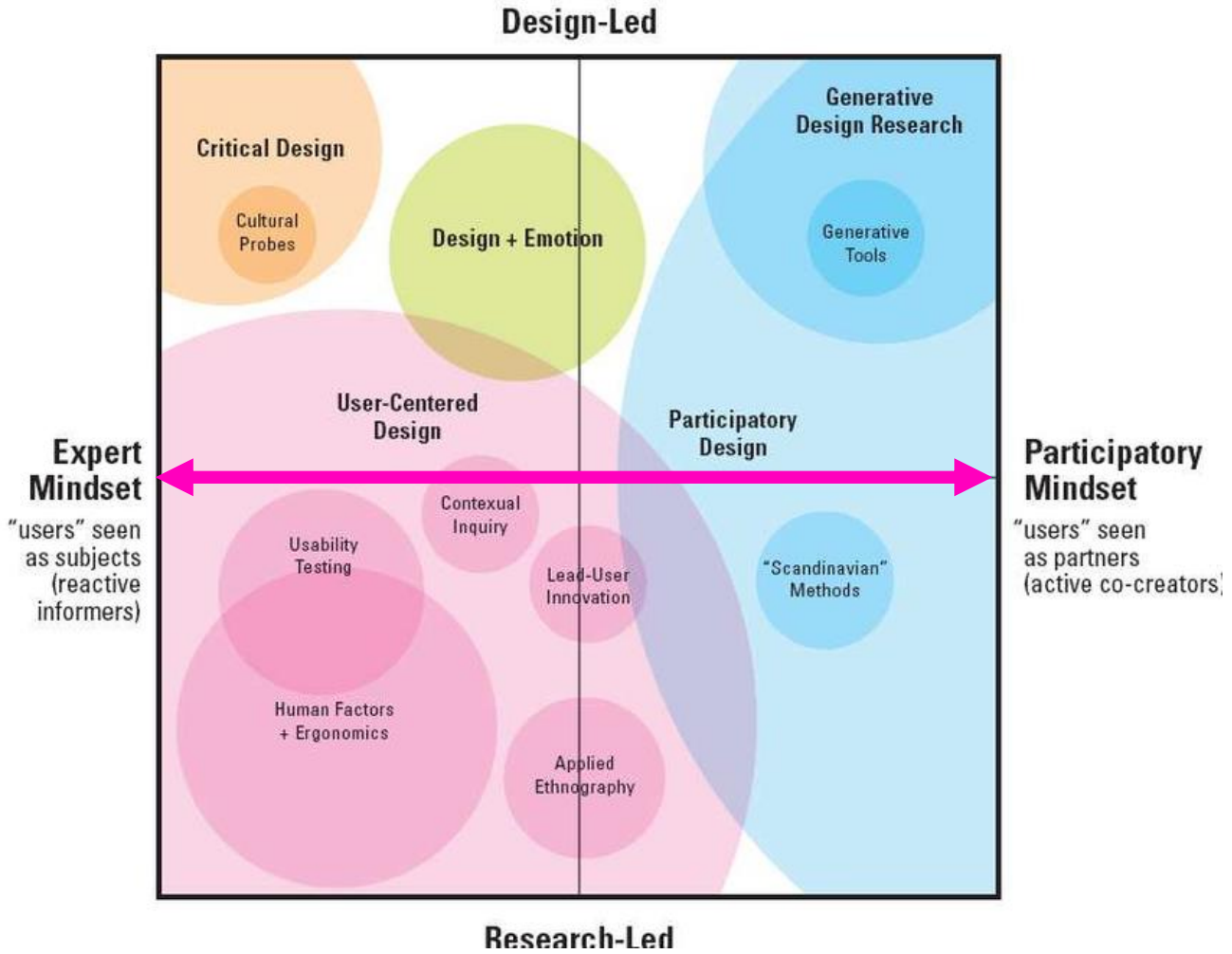
Mindsets

Expert Mindset:

- Designing FOR people
- People are referred as ‘subjects’, ‘users’, ‘consumers’, etc.

Participatory Mindset:

- Designing WITH people
- People as the true expert in areas of experience

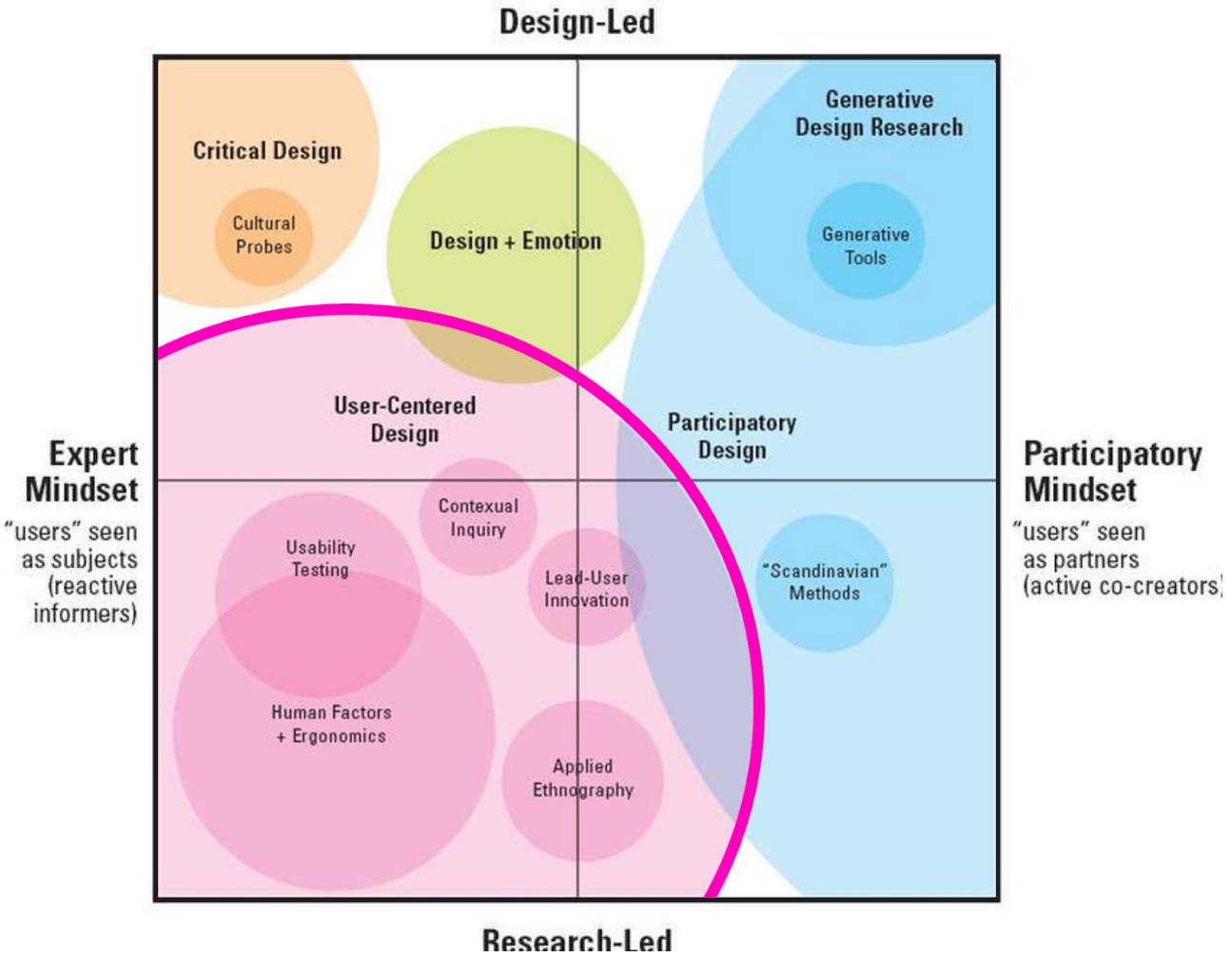


Sanders, E. B. N., & Stappers, P. J. (2012). Convivial toolbox: Generative research for the front end of design. Bis.

Design Research Landscape

User-Centered Design

- Helps make new products and services better meet the needs of ‘users’.
- Uses research-led approaches with an expert mindset to collect, analyze, and interpret data to develop specifications / principles to inform the design development.
- Tools and methods are applied in the evaluation of concepts and prototypes.

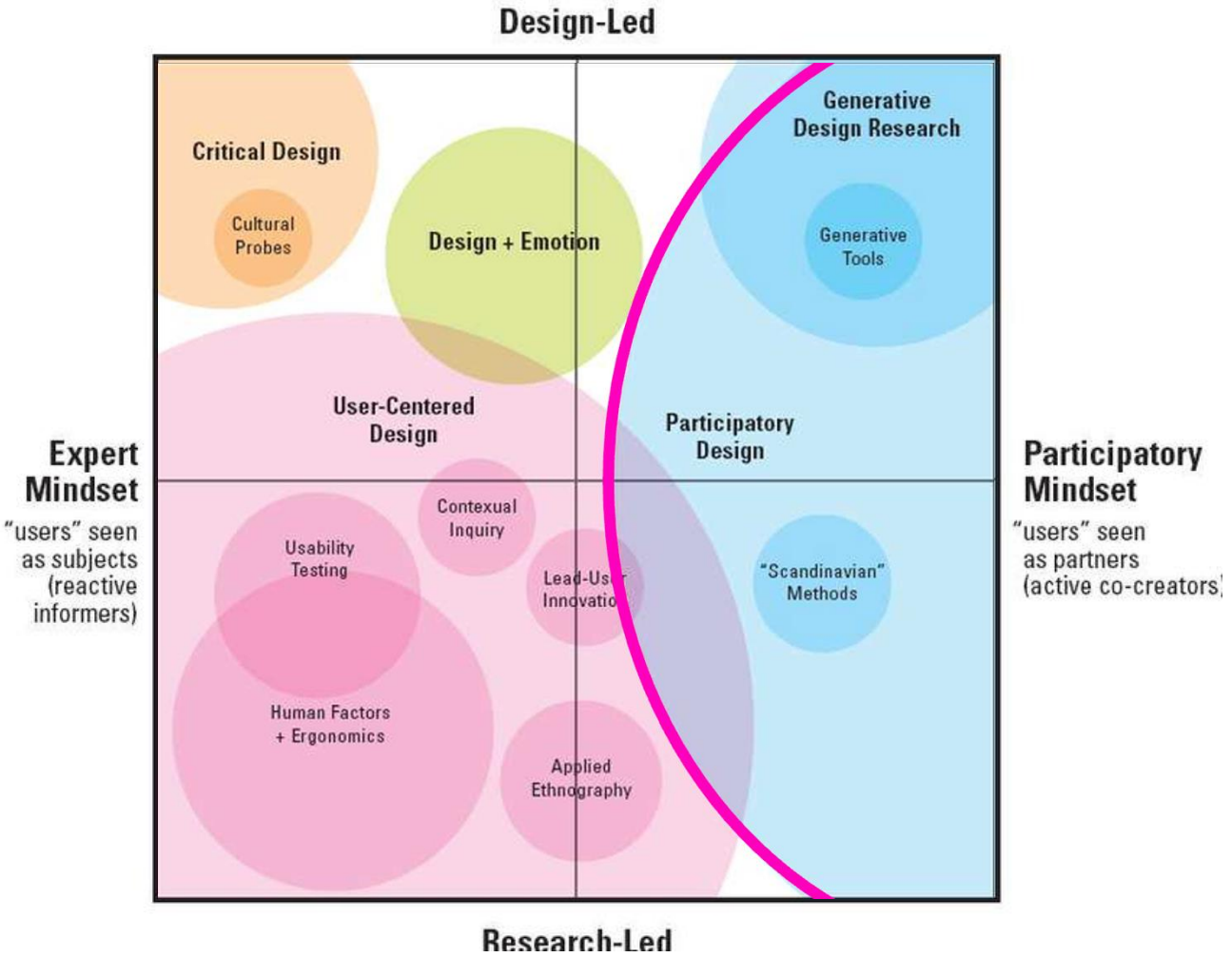


Sanders, E. B. N., & Stappers, P. J. (2012). Convivial toolbox: Generative research for the front end of design. Bis.

Design Research Landscape

Participatory Design

- Attempts to engage the target users in the design process to ensure the final product or service meets their needs.
- Focuses on involving the people who are and/or will become the 'users' throughout the design development process.
- Physical artifacts are used as thinking tools throughout the process.

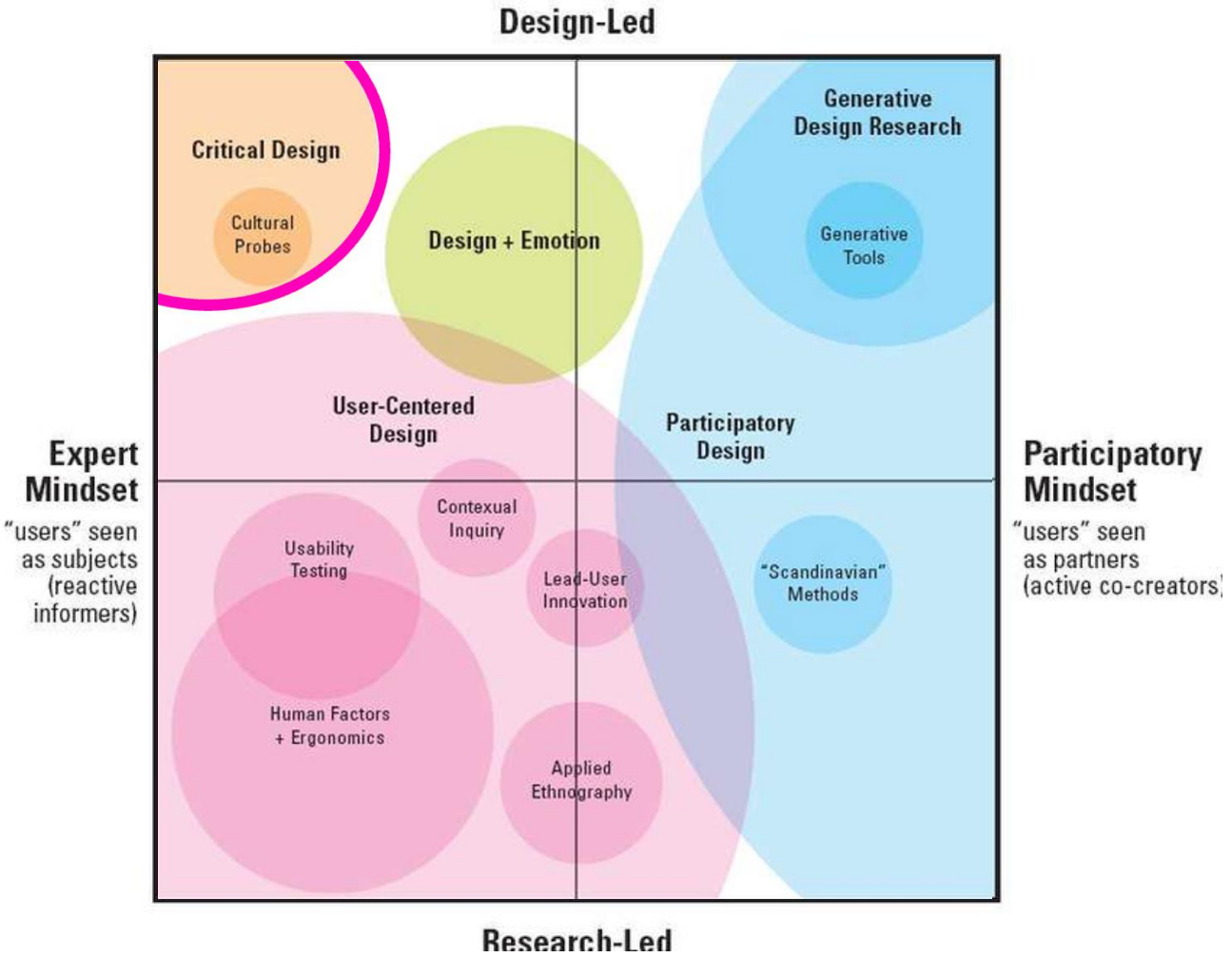


Sanders, E. B. N., & Stappers, P. J. (2012). Convivial toolbox: Generative research for the front end of design. Bis.

Design Research Landscape

Critical Design

- Emerged as a reaction to the user-centered approach, finding its focus on usability and utility pervasive.
- Evaluates the status quo and relies on design experts to create things that provoke reflection on the values people hold.
- Probes are intended to provide design inspiration, rather than to understand others' experiences.

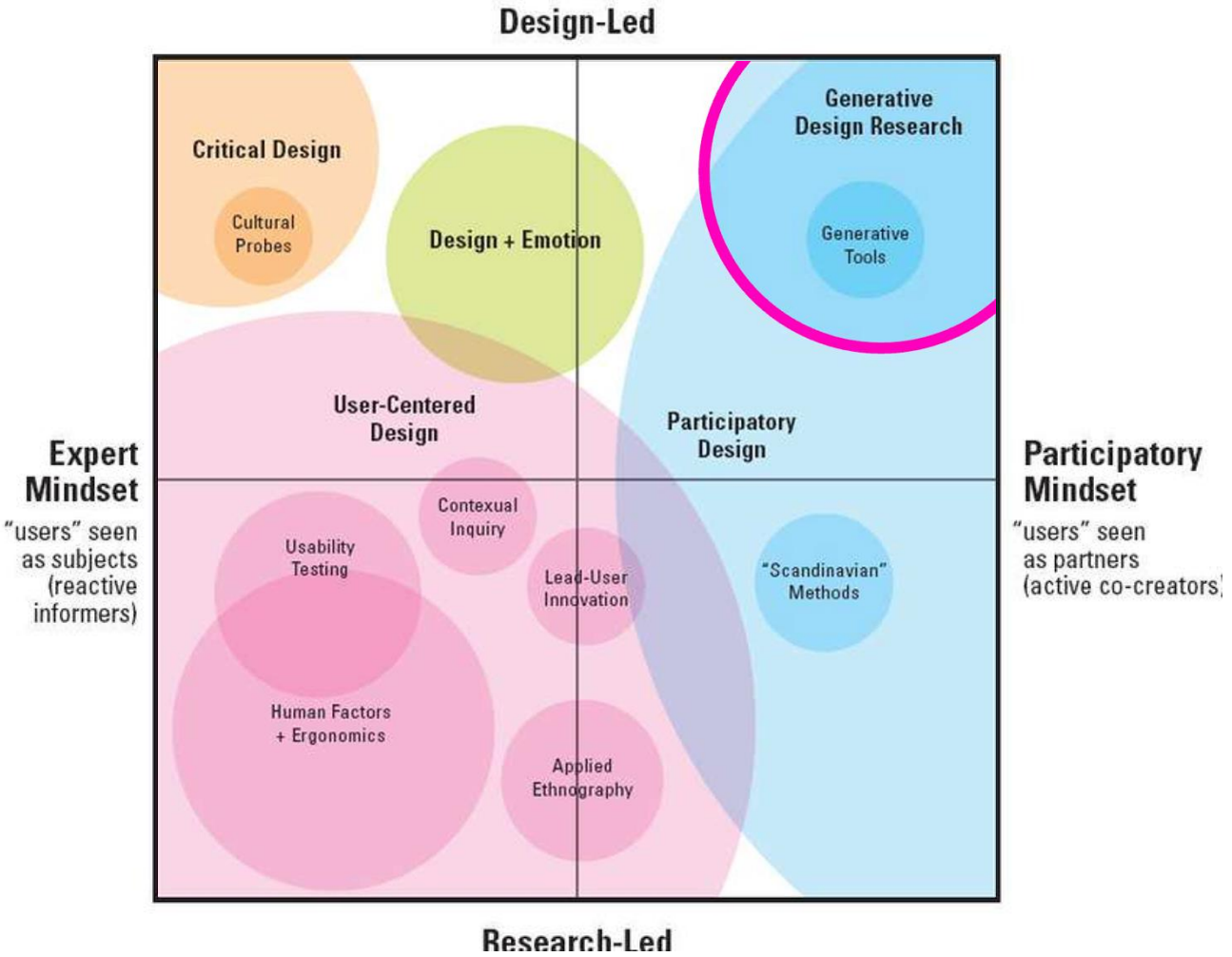


Sanders, E. B. N., & Stappers, P. J. (2012). Convivial toolbox: Generative research for the front end of design. Bis.

Design Research Landscape

Generative Design

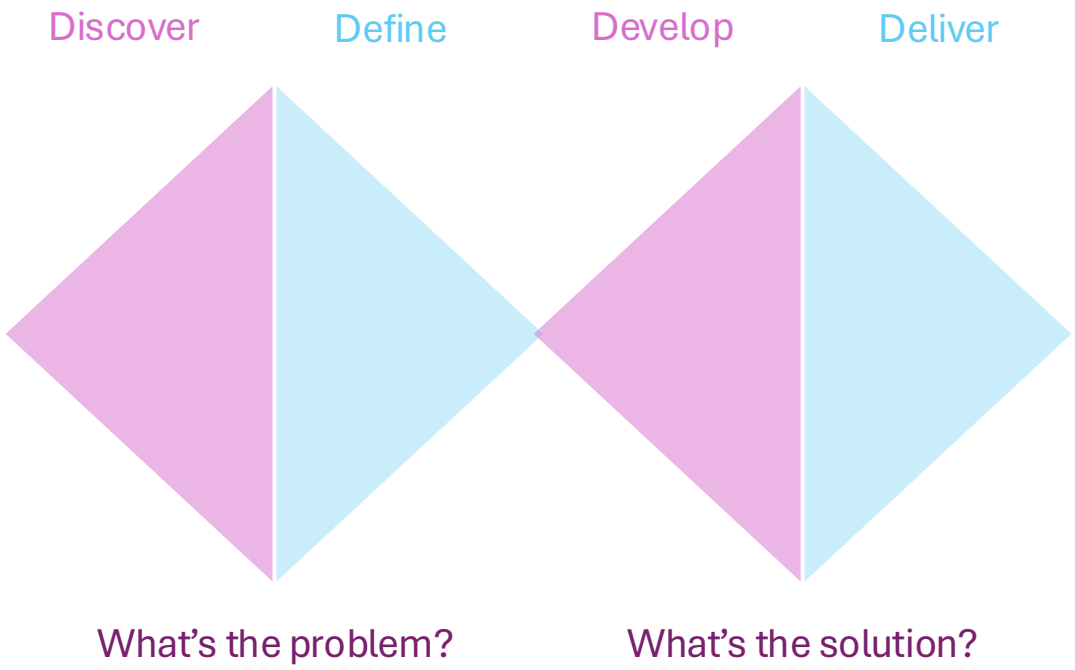
- Empowers everyday people to generate and promote alternatives to the current situation.
- Designers / researchers and stakeholders use a shared design language (visual and tangible) to communicate directly with each other.
- The design language is generative, allowing people to express countless ideas.



Sanders, E. B. N., & Stappers, P. J. (2012). Convivial toolbox: Generative research for the front end of design. Bis.

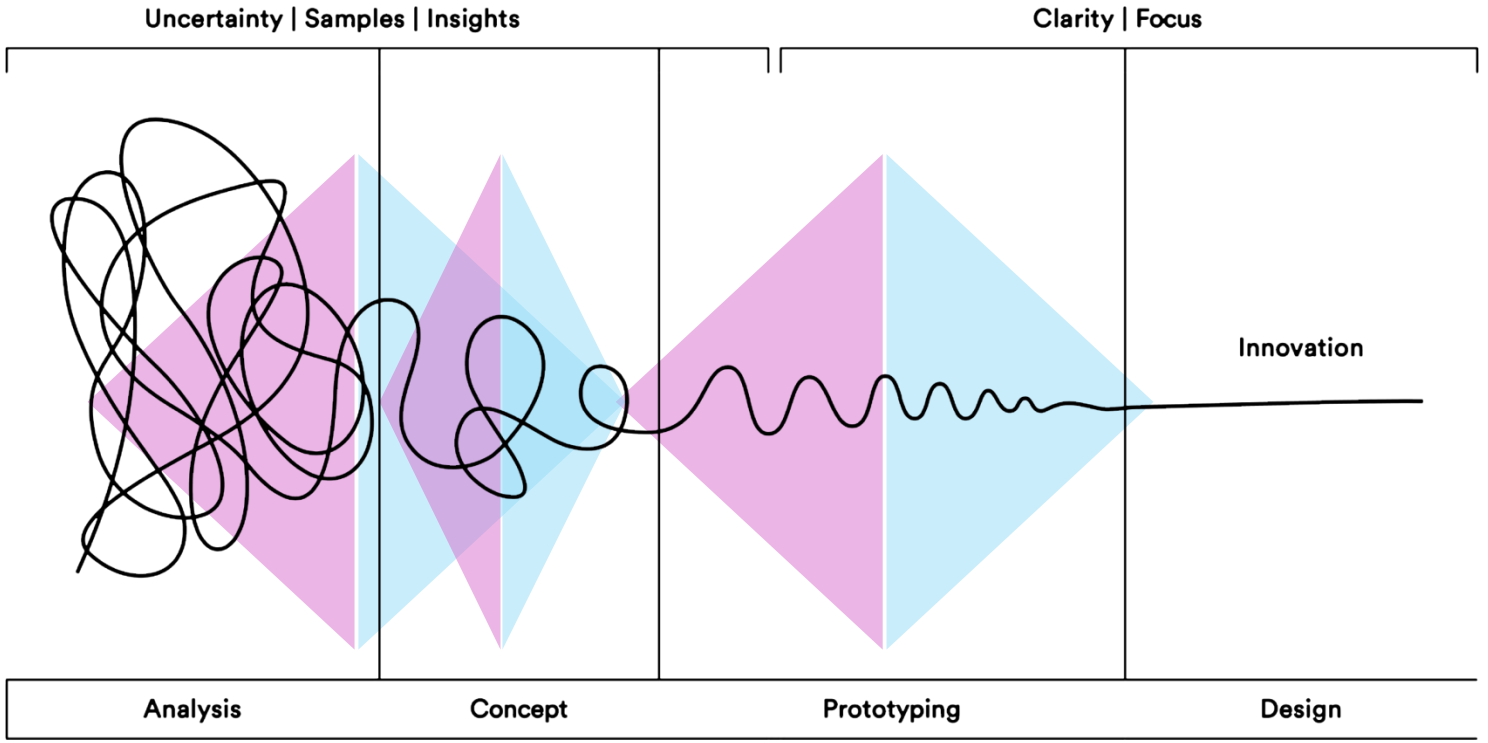
Design Development Process

- **Divergent thinking:** emphasizes broader, deeper, more original exploration, gathering more information, generating more ideas and thinking big
- **Convergent thinking:** prioritizes opportunities and emphasizes user needs and desires, editing down, making decisions, and driving towards an answer.



Design Process – Fuzzy Front End

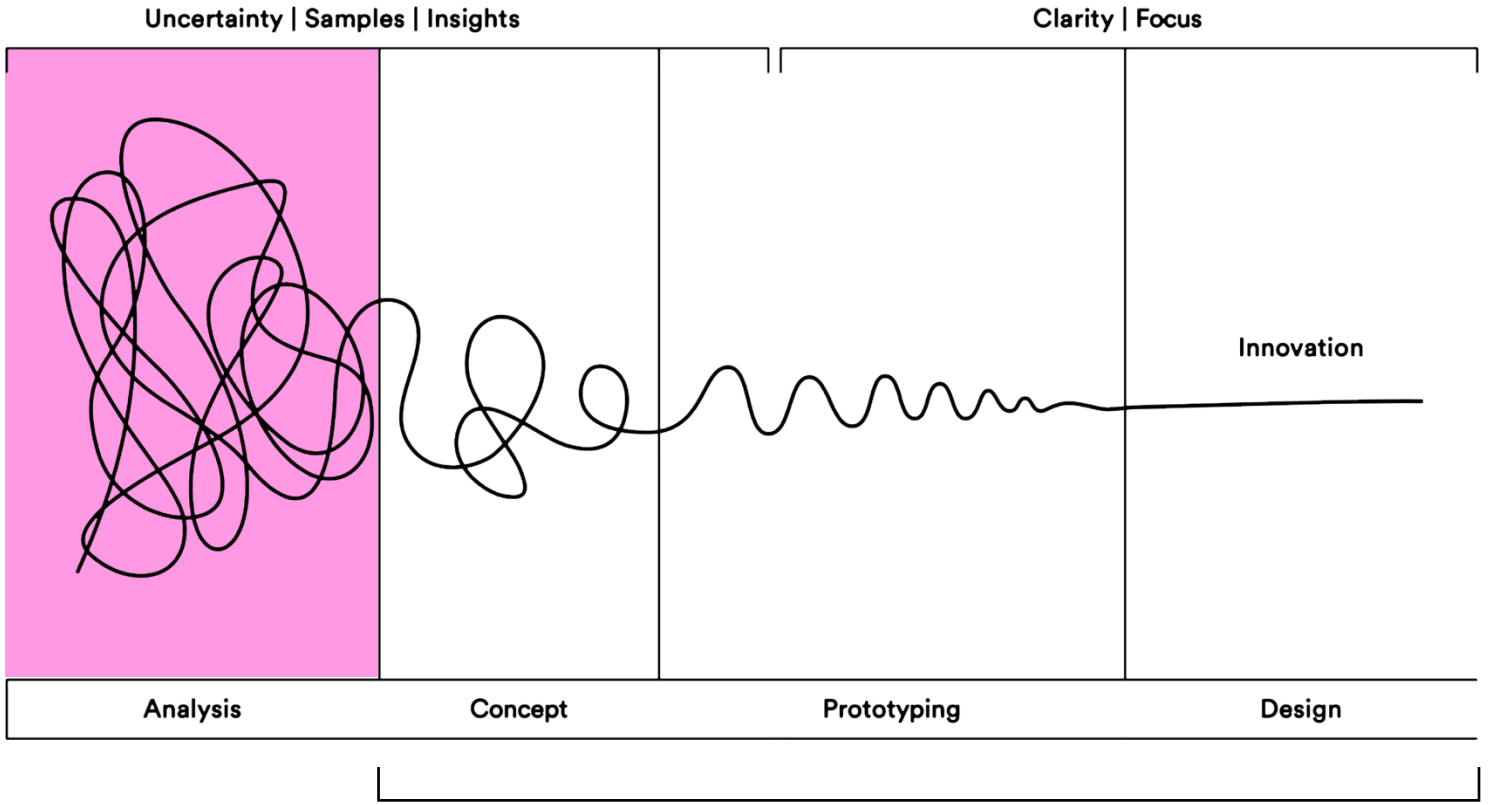
- The shape of the design development process has changed in response to the shifting foundations in the design research landscape..



Traditional Design Development Process

Design Process – Fuzzy Front End

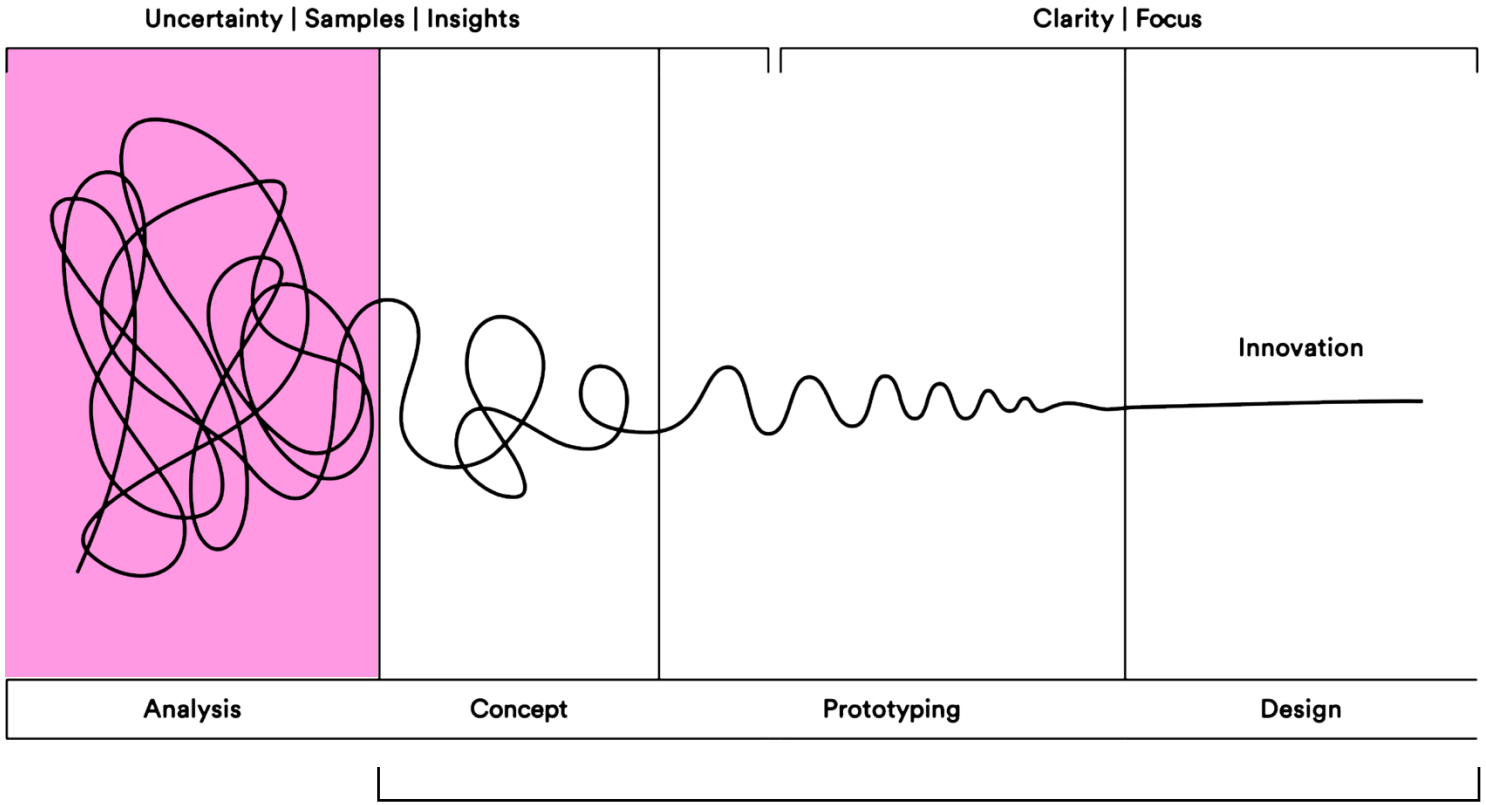
- The **‘fuzzy front end’** – the messy and chaotic phase – has been growing and gaining importance.
- In this pre-concept stage, ambiguity is highest, and the problem space itself is undefined.



Traditional Design Development Process

Design Process – Research

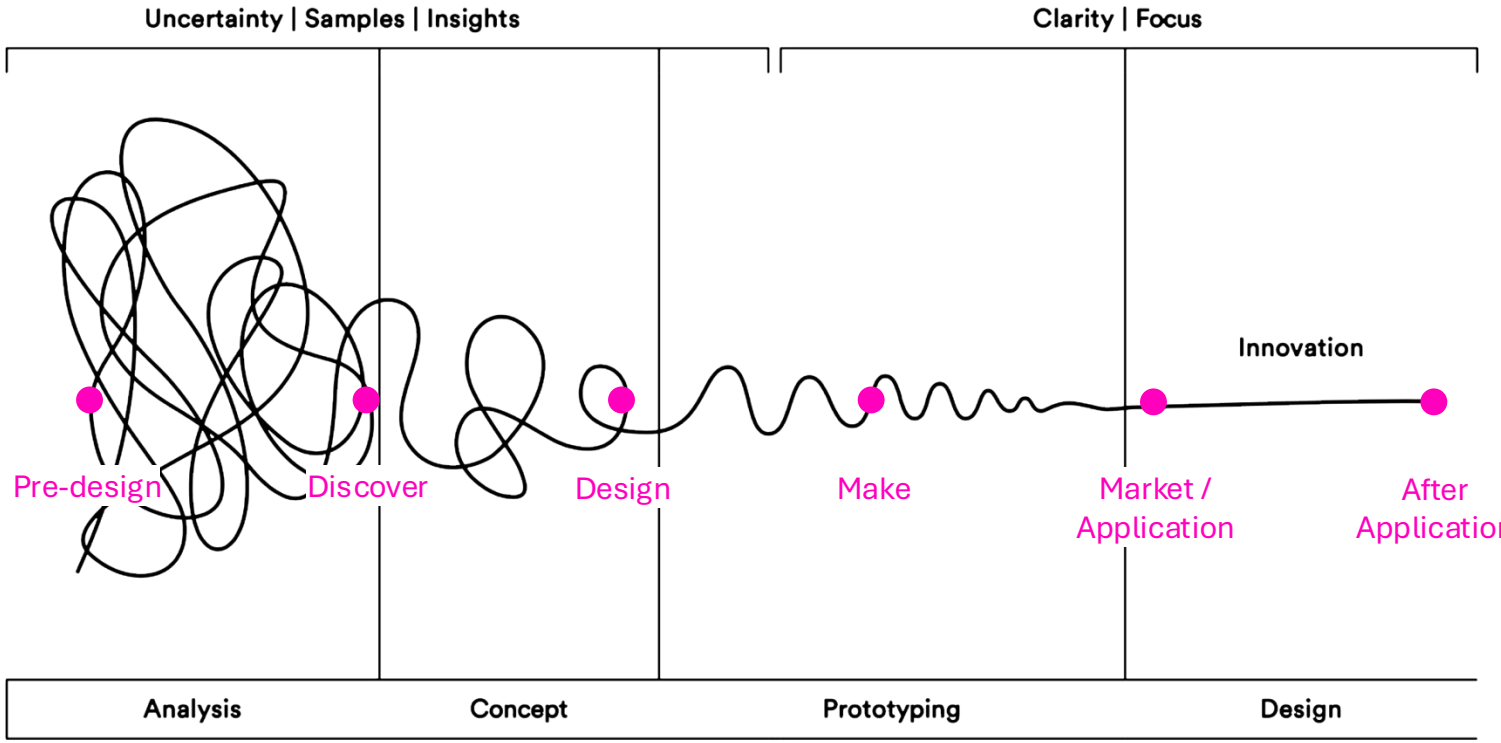
- Many activities take place here, focusing on informing and inspiring the exploration of open-ended questions and identifying the most relevant problems to address.
- Generative and exploratory design research happens here.
- The goal of the exploration is to define the fundamental problems and opportunities and to determine what could be designed.



Traditional Design Development Process

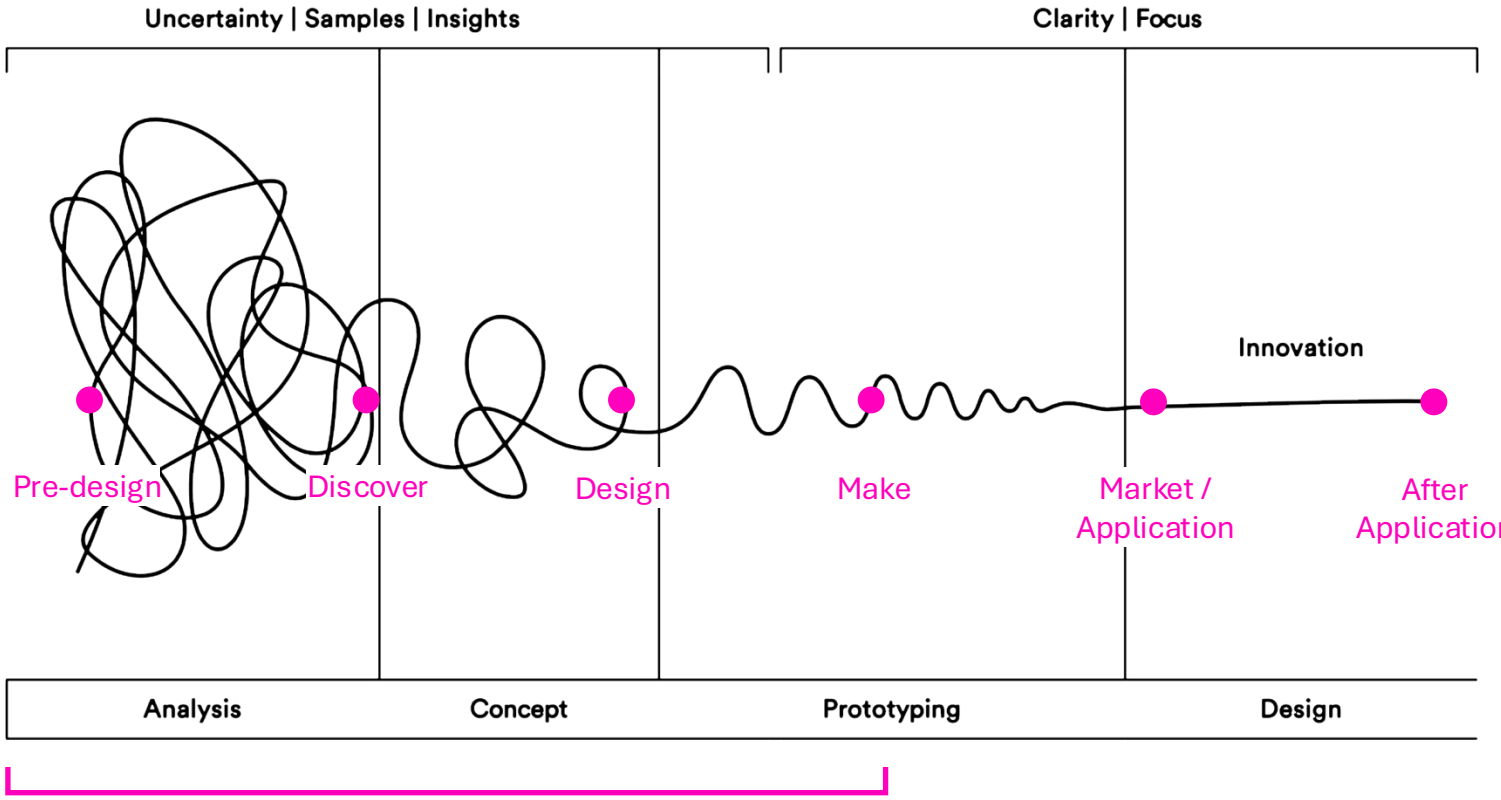
Design Development Process

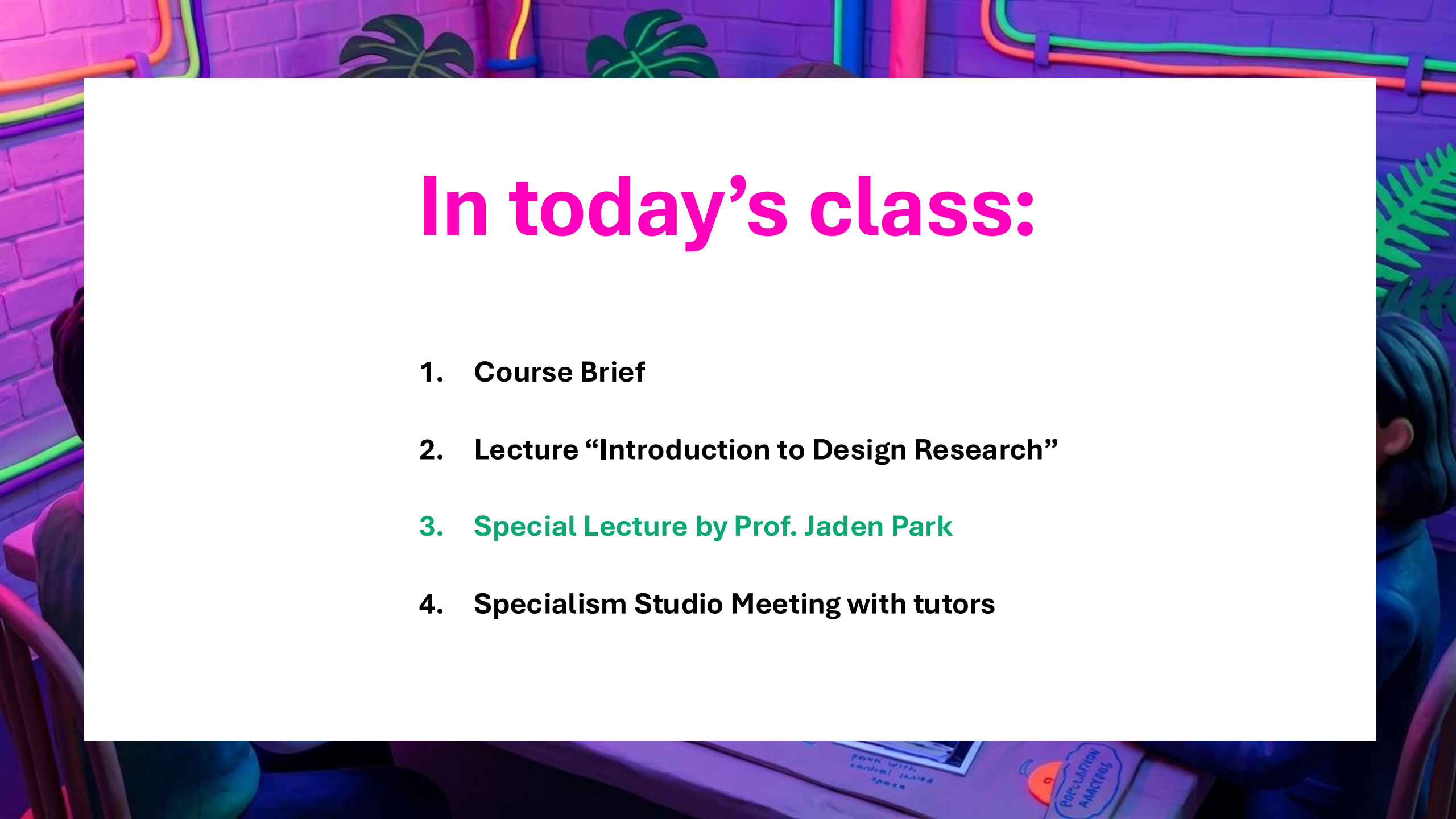
- **Pre-design:** where research, problem definition and the highest opportunity for innovation takes place.
- **Discover:** where opportunity identification and translation of the research to design occurs.
- **Design:** where continued exploration, design, and development takes place.
- Making
- Market
- After Application



Research as Design Process

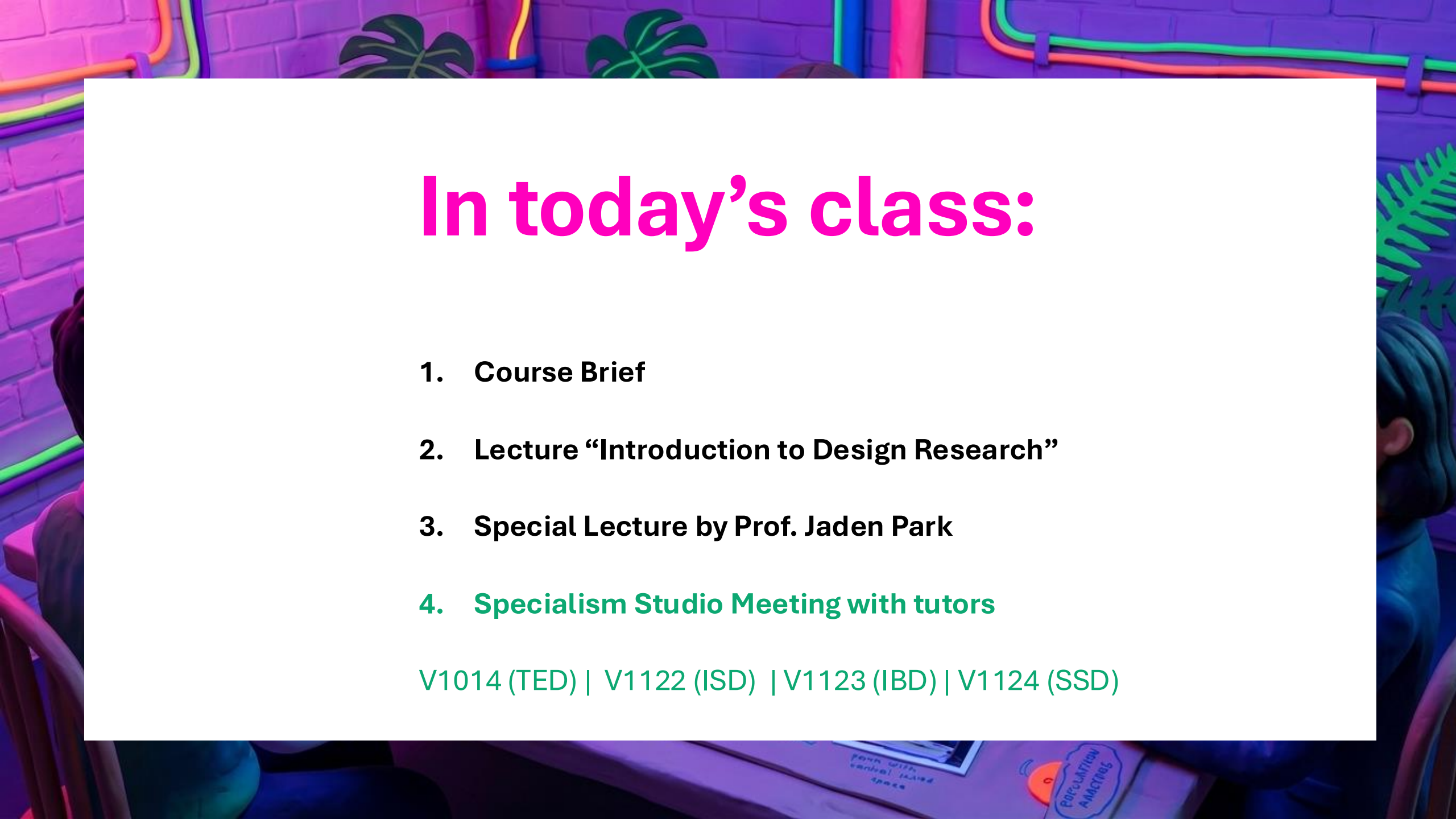
By focusing on ways of **expanding empirical observation** (gathering data) **and synthesizing evidence into insights** (framing questions and answers), we will experience how research is an integral part of designing.





In today’s class:

1. Course Brief
2. Lecture “Introduction to Design Research”
3. Special Lecture by Prof. Jaden Park
4. Specialism Studio Meeting with tutors



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V1014 (TED) | V1122 (ISD) | V1123 (IBD) | V1124 (SSD)



https://miro.com/app/board/uXjVHrjxrM0=/